

SMARTORDER-AI

TABLE OF CONTENT

Content	Page No.
CHAPTER 1 INTRODUCTION	
1.1 Introduction of the Project	
1.2 Purpose of the Project	
1.3 Function Requirements	
1.4 Problems in Existing System	
1.5 Main Modules	
CHAPTER 2 SYSTEM REQUIREMENTS	
2.1 Hardware & Software Requirements	
2.1.1 Server Side Requirements	
2.1.2 Developer Side Requirements	
2.1.3 User Side Requirement	
CHAPTER 3 SYSTEM ANALYSIS AND DESIGN	
3.1 System Analysis and Design	
3.2 System Development Life Cycle (SDLC)	
3.2.1 Needs of SDLC	
3.2.2 SDLC Cycle	
3.3 Spiral Model	
3.4 Feasibility Study	
CHAPTER 4 FRONT END OF SYSTEM	
4.1 About Front End	
4.1.1 about react.js and Taiwind css	
4.1.2 about HTML	
4.1.3 about CSS	
4.1.4 about JavaScript	
4.2 Why Use react.js	
CHAPTER 5 BACK END OF SYSTEM	
5.1 About Back End	
5.1.1 about node.js and mongodb	
5.1.2 How mongodb Works	
5.1.3 mongodb Features	
5.2 Why use mongodb and node.js	
CHAPTER 6 SYSTEM DESIGN	
6.1 Data Flow Diagram	
6.2 Use Case Diagram	

6.3 Activity Diagram

6.4 E-R Diagram

CHAPTER 7 DATA DICTIONARY

7.1 Introduction

7.2 List of Tables

7.2.1: Login

7.2.2: State

7.2.3: City

7.2.4: Registration

7.2.5: T-Shirt Booking Table

7.2. 6: SmartOrder AI Table

7.2.7: Inquiry

7.2.8: Feedback

CHAPTER 8 TESTING

8.1 Testing Plan

8.2 Testing Strategies

8.3 Testing Method

8.4 Test Case

CHAPTER 9 SSNAPSHOT OF WEBSITE

9.1 Admin Site

9.2 Users Site

CHAPTER 10 ADVANTAGES

10.1 Advantages

10.2 Limitations

CHAPTER 11 CONCLUSION AND FUTURE ENHANCEMENT

11.1 Conclusion

11.2 Future Enhancement

CHAPTER 12 BIBLIOGRAPHY

12.1 Web Reference

CHAPTER 1

INTRODUCTION

1.1 Introduction of the Project

- The **SmartOrder AI** project is an intelligent e-commerce web application that integrates **Artificial Intelligence based decision analytics** with a traditional online shopping platform. The main purpose of this system is to analyze user behavior during shopping and classify purchase decisions into **Rational Purchase** or **Impulse Purchase**.
- The platform allows users to browse products, view product details, add items to the cart, and complete purchases. While users interact with the platform, the system records various behavioral metrics such as:
 - Time spent on product pages
 - User interaction frequency
 - Cart activity
 - Decision time
- These behavioral signals are processed by the AI decision system to generate a **Decision Score**, which determines whether a purchase was impulsive or rational.
- The system also provides an **Admin Control Hub** where administrators can:
 - Manage products
 - View payment records
 - Track orders
 - Monitor AI behavior logs
 - Analyze user decision patterns
 - View analytics dashboards

1.2 Purpose of the Project

- The main purpose of the SmartOrder AI project is to develop an **intelligent e-commerce system that can analyze customer behavior and purchasing decisions in real time**.
- Traditional e-commerce websites focus only on selling products, but they do not analyze **why users purchase certain products quickly or hesitate before buying**.
- This project introduces an AI-based decision model that helps to:
 - Detect impulse buying behavior
 - Analyze customer engagement
 - Improve product decision analytics
 - Provide real-time behavioral insights
 - Help administrators understand customer purchase patterns
- The system helps both customers and businesses:
- For Customers
 - Better shopping experience
 - Improved product discovery

- Reduced impulse purchases
- For Businesses
- Customer behavior analytics
- Decision pattern analysis
- Improved marketing strategies
- Data-driven decision making

1.3 Function Requirements

- **Functional requirements describe the features and operations that the system must perform.**
- **The SmartOrder AI platform includes the following functional requirements.**

- **1 User Authentication**

The system must allow users to:

- **Register an account**
- **Login to the system**
- **Logout securely**

- **2 Product Browsing**

Users should be able to:

- **View available products**
- **Filter products by color or category**
- **View product details**
- **Compare products**

- **3 Cart Management**

Users must be able to:

- **Add products to cart**
- **Remove products**
- **Update quantity**
- **View cart total**

- **4 Checkout System**

The system must support checkout functionality where users can:

- **Enter shipping information**
- **Select payment method**
- **Place orders**

- **5 AI Behavior Tracking**

The system records user interaction data such as:

- **Time spent on product page**

- **Add to cart speed**
- **Scroll behavior**
- **Product views**

These signals are used to generate a Decision Score.

- **6 AI Decision Classification**

The system classifies purchases into:

- **Rational Purchase**
- **Impulse Purchase**

based on the AI scoring model.

- **7 Admin Control Panel**

The admin panel allows administrators to manage system data.

Admin features include:

- **Product Management**
- **Payment Records**
- **Orders Records**
- **AI Behavior Logs**
- **Analytics Dashboard**
- **Contact Messages**
- **User Management**

1.4 Problems in Existing System

- Existing e-commerce systems primarily focus on product sales and do not analyze customer purchasing behavior in depth. These systems lack intelligent mechanisms to identify whether a purchase was impulsive or rational.
- Some major problems in traditional e-commerce platforms include:
 - **1 No Behavioral Intelligence**
 - Most platforms do not track user behavior such as decision time, interaction patterns, or hesitation during purchase.
 - **2 No Decision Analysis**
 - Traditional systems cannot determine whether a customer made a rational purchase or an impulse purchase.
 - **3 Limited Analytics**

Existing systems provide only basic statistics such as:

 - Total sales
 - Number of orders
 - Revenue reports
- They do not provide insights into **why users purchase certain products.**

- **4 No AI Integration**
 - Many systems lack AI models that can analyze real-time user interactions and generate decision insights.
- **5 Limited Admin Insights**

Administrators cannot track behavioral patterns such as:

 - hesitation
 - product engagement
 - impulse purchases

The SmartOrder AI system solves these problems by integrating **behavioral analytics and AI decision intelligence**.

1.5 Main Modules

1 User Module

- This module allows users to interact with the system.

Features include:

- User Registration
- User Login
- Product Browsing
- Add to Cart
- Checkout
- Order Placement

2 AI Behavior Analysis Module

- This module tracks and analyzes user interaction with products.

It collects data such as:

- Product views
- Time spent
- Add to cart time
- Interaction events

This information is used to generate **AI Decision Scores**.

3 Decision Engine Module

- This module calculates decision scores based on behavioral parameters.

The system classifies purchase decisions into:

- Rational Purchase

4 Admin Management Module

- This module allows administrators to manage the entire platform.

Features include:

- Add / Edit / Delete Products
- Manage Orders
- View Payment Records
- View AI Logs
- Manage Users
- Manage Contact Messages

5 Analytics Module

- This module provides graphical analytics and reports.

It includes:

- Rational vs Impulse Purchase Graph
- Revenue Trend
- Product Popularity
- User Time Spent Analysis
- AI Score Trends

CHAPTER 2

SYSTEM REQUIREMENTS

2.1 Hardware & Software Requirements

❖ SYSTEM REQUIREMENTS

- The **System Requirements** describe the hardware and software environment required to develop, run, and maintain the SmartOrder AI system.
- This chapter explains the technical requirements needed for:
 - Server-side execution
 - Development environment
 - User-side access
- The SmartOrder AI platform is a **web-based intelligent e-commerce system**, therefore it requires both backend server support and a frontend browser interface.
- The system is built using **modern full-stack technologies** including:
 - React.js for frontend
 - Node.js and Express.js for backend
 - MongoDB for database management
 - Tailwind CSS for responsive UI design
- The application runs on a **local development server during development** and can be deployed to cloud platforms for production.

❖ Hardware Requirement

- Minimum hardware requirements for development and testing:
 - Processor
Intel Core i3 or higher
 - RAM
4 GB minimum (8 GB recommended)
 - Storage
Minimum 10 GB free disk space
 - Display
1366 × 768 resolution or higher
 - Internet Connection
Required for package installation and API communication

❖ Software Requirements

- The following software technologies are required for development and execution.
 - Operating System
Windows 10 / Windows 11 / Linux / macOS
 - Frontend Technologies
HTML5
CSS3
Tailwind CSS
JavaScript
React.js
 - Backend Technologies
Node.js
Express.js
 - Database
MongoDB
 - Development Tools
Visual Studio Code
Git
Node Package Manager (NPM)
 - Browser Support
Google Chrome
Microsoft Edge
Mozilla Firefox

2.1.1 Server Side Requirements

- Server Software
Node.js Runtime Environment
- Backend Framework
Express.js
- Database Server
MongoDB

- API Communication
REST API
- Server Port
Typically runs on port **5000**

2.1.2 Developer Side Requirements

- Code Editor
Visual Studio Code
- Version Control
Git
- Package Manager
Node Package Manager (NPM)
- Frontend Framework
React.js
- Backend Runtime
Node.js
- Database
MongoDB

2.1.3 User Side Requirements

- Device
Laptop, Desktop, Tablet, or Smartphone
- Internet Connection
Stable internet connection required
- Web Browser
Google Chrome
Mozilla Firefox
Microsoft Edge

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis and Design

❖ Systems Analysis

- It is a process of collecting and interpreting facts, identifying the problems, and decomposition of a system into its components.
- In the **SmartOrder AI project**, system analysis was performed to understand how users interact with the e-commerce platform and how their behavior can be analyzed using AI.
- The system design defines how the application is structured and how its modules communicate with each other.
- The SmartOrder AI system includes several integrated components such as:
 - User Interface (Frontend)
 - Backend Server
 - Database Management
 - AI Behavior Analysis Engine
 - Admin Dashboard
- System analysis is conducted for the purpose of studying a system or its parts in order to identify its objectives.
- It is a problem solving technique that improves the system and ensures that all the components of the system work efficiently to accomplish their purpose.
- Analysis specifies what the system should do.
- A system must have three basic constraints –
- A system must have some structure and behavior which is designed to achieve a predefined objective.
- Interconnectivity and interdependence must exist among the system components.
- The objectives of the organization have a higher priority than the objectives of its subsystems.

❖ Systems Design

- It is a process of planning a new business system or replacing an existing system by defining its components or modules to satisfy the specific requirements. Before planning, you need to understand the old system thoroughly and determine how computers can best be used in order to operate efficiently.
- System Design focuses on how to accomplish the objective of the system.
- System Analysis and Design (SAD) mainly focuses on –
 - Systems
 - Processes

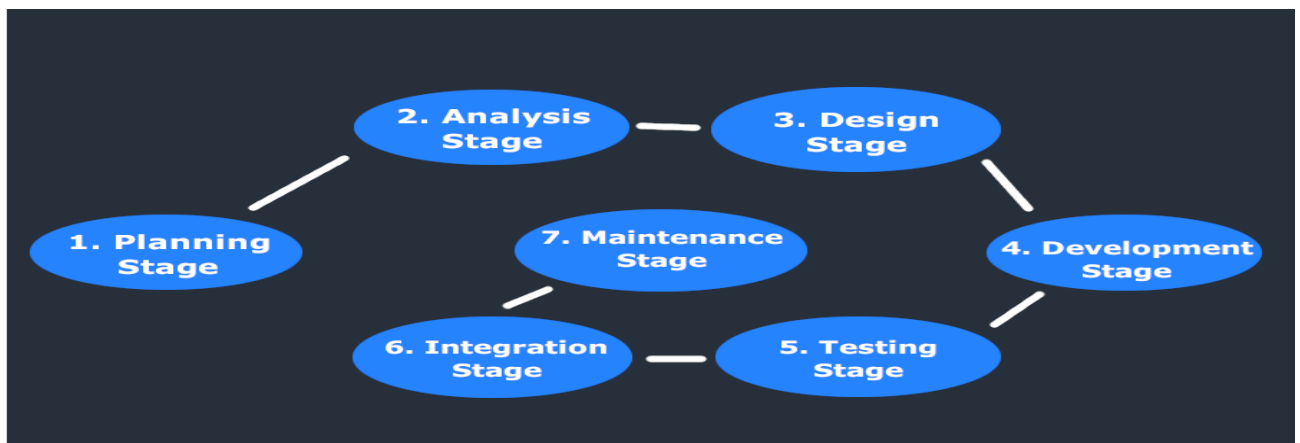
- Technology

❖ System Workflow



- The workflow of the SmartOrder AI system is as follows:
 - User visits the SmartOrder AI website.
 - The system displays available products.
 - User views product details and interacts with the platform.
 - The system records user interaction behavior.
 - User adds products to the cart.
 - At checkout, the AI decision system analyzes user behavior.
 - The system generates a **Decision Score**.
 - The purchase is classified as **Rational Purchase or Impulse Purchase**.
 - The admin dashboard displays analytics and reports.

3.2 System Development Life Cycle (SDLC)



- The software development process is normally long and tedious.
- But project managers and system analysts can leverage software development life cycles to outline, design, develop, test, and eventually deploy information systems or software products with greater regularity, efficiency, and overall quality.

❖ System Development Life Cycle

- A system development life cycle or SDLC is essentially a project management model.
- It defines different stages that are necessary to bring a project from its initial idea or conception all the way to deployment and later maintenance.

❖ 7 Stages of the System Development Life Cycle

- There are seven primary stages of the modern system development life cycle. Here's a brief breakdown:
- Planning Stage
- Feasibility or Requirements of Analysis Stage
- Design and Prototyping Stage
- Software Development Stage
- Software Testing Stage
- Implementation and Integration
- Operations and Maintenance Stage

❖ Planning Stage



- Before we even begin with the planning stage, the best tip we can give you is to take time and acquire proper understanding of app development life cycle.
- It helps to define the problem and scope of any existing systems, as well as determine the objectives for their new systems.
- By developing an effective outline for the upcoming development cycle, they'll theoretically catch problems before they affect development.
- Perhaps most importantly, the planning stage sets the project schedule, which can be of key importance if development is for a commercial product that must be sent to market by a certain time.

❖ Analysis Stage

- The analysis stage includes gathering all the specific details required for a new system as well as determining the first ideas for prototypes.
 - Requirements included:
 - User login and registration
 - Product browsing system
 - Cart management
 - Checkout process
 - AI decision analysis
 - Admin dashboard
 - Analytics and reports
- Developers may:
- Perform research and analysis to determine the needs of end-users

- Furthermore, developers will often create a software requirement specification or SRS document.
- This includes all the specifications for software, hardware, and network requirements for the system they plan to build. This will prevent them from overdrawing funding or resources when working at the same place as other development teams.

❖ Design Stage



- The design stage is a necessary precursor to the main developer stage.
- Developers will first outline the details for the overall application, alongside specific aspects, such as its:
 - Design decisions included:
 - Frontend UI layout
 - Backend API structure
 - Database schema
 - AI scoring logic
 - Admin dashboard interface
- System interfaces
- Network and network requirements
- Databases
- Operation, training, and maintenance plans will all be drawn up so that developers know what they need to do throughout every stage of the cycle moving forward.

❖ Development Stage

- In the development phase, the system was implemented using modern web technologies.
 - Frontend development included:
 - Creating responsive UI components
 - Product display pages
 - Cart functionality
 - Checkout page
 - Backend development included:
 - API creation
 - Database integration
 - AI behavior tracking

- Admin panel functionality

- The development stage is the part where developers actually write code and build the application according to the earlier design documents and outlined specifications.
- Programming languages can include staples such as reactjs, nodejs javascript, and more.
- Developers will choose the right programming code to use based on the project specifications and requirements.

❖ Testing Stage



- Building software is not the end.
- Now it must be tested to make sure that there aren't any bugs and that the end-user experience will not negatively be affected at any point.
- During the testing stage, developers will go over their software with a fine-tooth comb, noting any bugs or defects that need to be tracked, fixed, and later retested.
- Depending on the skill of the developers, the complexity of the software, and the requirements for the end-user, testing can either be an extremely short phase or take a very long time.

❖ Implementation and Integration Stage

- After testing, the overall design for the software will come together. Different modules or designs will be integrated into the primary source code through developer efforts, usually by leveraging training environments to detect further errors or defects.
- The information system will be integrated into its environment and eventually installed. After passing this stage, the software is theoretically ready for market and may be provided to any end-users.

❖ Maintenance Stage



- The SDLC doesn't end when software reaches the market. Developers must now move into a maintenance mode and begin practicing any activities required to handle issues reported by end-users.
- This can include handling residual bugs that were not able to be patched before launch or resolving new issues that crop up due to user reports. Larger systems may require longer maintenance stages compared to smaller systems.

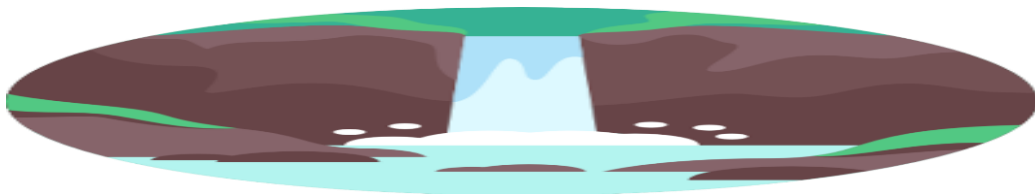
❖ Role of System Analyst

- An SDLC's system analyst is, in some ways, an overseer for the entire system. They should be totally aware of the system and all its moving parts and can help guide the project by giving appropriate directions.
- The system analyst should be:
- A good planner so that development tasks can be carried out on time at each phase of the development cycle
- Thus, systems analysts should have an even mix of interpersonal, technical, management, and analytical skills altogether. They're versatile professionals that can make or break an SDLC.

❖ 6 Basic SDLC Methodologies

- Although the system development life cycle is a project management model in the broad sense, six more specific methodologies can be leveraged to achieve specific results or provide the greater SDLC with different attributes.

❖ Waterfall Model



- The waterfall model is the oldest of all SDLC methodologies. It's linear and straightforward and requires development teams to finish one phase of the project completely before moving on to the next.
- Each stage has a separate project plan and takes information from the previous stage to avoid similar issues (if encountered). However, it is vulnerable to early delays and can lead to big problems arising for development teams later down the road.

❖ Iterative Model

- The iterative model focuses on repetition and repeat testing. New versions of a software project are produced at the end of each phase to catch potential errors and allow developers to constantly improve the end product by the time it is ready for market.
- One of the upsides to this model is that developers can create a working version of the project relatively early in their development life cycle, so implement the changes are often less expensive.

❖ Spiral Model



- Spiral models are flexible compared to other methodologies. Projects pass through four main phases again and again in a metaphorically spiral motion.
- It's advantageous for large projects since development teams can create very customized products and incorporate any received feedback relatively early in the life cycle.

❖ Agile Model



- The agile model is relatively well-known, particularly in the software development industry.
- The agile methodology prioritizes fast and ongoing release cycles, utilizing small but incremental changes between releases. This results in more iterations and many more tests compared to other models.

3.2.1 Needs of SDLC

- Software Development Life Cycle encourages the software development team to develop a software product in a disciplined and systematic way.
- As the software product is developed in a team, it is necessary to have an ideal way to follow for successful completion of software development.
- If there will not be any development process, every developer will think of their own working freedom that will create chaos.
- It will be very difficult to deliver a quality product in a specified time.
- SDLC sets the series of stages to follow that not only complete the development properly but also consume less time in the development process.
- It provides a quality software solution within specified time and expense.
- Use of System Requirement Specification (SRS) document as a guideline of the project leads to client satisfaction.
- For the SmartOrder AI project, SDLC ensured that the system was developed in a structured and organized manner.

3.2.2 SDLC Cycle

- The systems development life cycle (SDLC) is a conceptual model used in project management that describes the stages involved in an information system development

project, from an initial feasibility study through maintenance of the completed application.

- SDLC can apply to technical and non-technical systems. In most use cases, a system is an IT technology such as hardware and software.
- Project and program managers typically take part in SDLC, along with system and software engineers, development teams and end-users.
- Every hardware or software system will go through a development process which can be thought as an iterative process with multiple steps.
- SDLC is also an abbreviation for Synchronous Data Link Control and software development life cycle.
- Software development life cycle is a very similar process to systems development life cycle, but it focuses exclusively on the development life cycle of software.

3.3 Spiral Model

- **Definition:** The spiral model is similar to the incremental development for a system, with more emphasis placed on risk analysis.
- The spiral model has four phases: Planning, Design, Construct and Evaluation. A software project repeatedly passes through these phases in iterations (called Spirals in this model).
- **Description:** These phases are - Planning: This phase starts with the gathering of business requirements.
- In the subsequent spirals as the product matures, identification of system requirements and unit requirements are done in this phase.
- **Design:** Design phase starts with the design in the baseline spiral and involves architectural, logical design of modules, physical product design and final design in the successive spirals.
- Then in the successive spirals with higher clarity on requirements and design a working model of the software called build is developed with a version number.
- In the SmartOrder AI project, the spiral model concept helps in improving system modules gradually by analyzing user behavior and updating the AI system.

3.4 Feasibility Study

- The feasibility study is an evaluation and analysis of the potential of the proposed project which is based on extensive investigation and research to support the process of decision making.

- For the SmartOrder AI project, several types of feasibility were considered.
- Feasibility is the measure of how beneficial the development of information system will be to an organization.
- The feasibility analysis is categorized under four different types.

- 1) Operational Feasibility
- 2) Technical Feasibility
- 3) Schedule Feasibility
- 4) Economic Feasibility

1) Operational Feasibility:

- Operational feasibility evaluates whether the system will function effectively in a real environment.
- The SmartOrder AI platform provides a user-friendly interface and simple navigation, making it easy for users and administrators to operate.

2) Technical Feasibility:

- Technical feasibility determines whether the required technology and tools are available.
 - The SmartOrder AI system uses widely available technologies:
 - React.js
 - Node.js
 - MongoDB
 - Tailwind CSS
- These technologies are open-source and widely supported, making the project technically feasible.

3. Schedule Feasibility:

- Schedule feasibility determines whether the project can be completed within the given time frame.

- The SmartOrder AI project was developed using a structured development plan and completed within the allocated development period.

4. Economic Feasibility

- Economic feasibility analyzes the cost of development.
- Since the system uses open-source technologies, the development cost is minimal.
- Required resources include:
 - Developer time
 - Computer system
 - Internet connection
- Therefore, the project is economically feasible.

CHAPTER 4

FRONT END OF SYSTEM

4.1 About Front End

- The **Front End** of a system is the user-facing part of the application that users interact with directly. It includes the visual interface, layout, navigation, and interactive elements of the website.
- In the **SmartOrder AI project**, the front end is responsible for displaying product information, collecting user inputs, and communicating with the backend server.
- The frontend provides features such as:
 - Product browsing
 - Product detail viewing
 - Add to cart functionality
 - Checkout system
 - User login and registration
 - Admin dashboard interface
- The SmartOrder AI frontend is developed using modern web technologies including:
 - **React.js**
 - **Tailwind CSS**
 - **HTML**
 - **CSS**
 - **JavaScript**
- These technologies help create a responsive, fast, and interactive user interface.

4.1.1 About React.js and Tailwind CSS

- **React.js**
- React.js is a popular **JavaScript library used for building dynamic user interfaces**, especially for single-page applications.
- React allows developers to create reusable components that make the application more organized and easier to maintain.
- Key features of React.js include:
 - Component-based architecture
 - Fast rendering using Virtual DOM
 - Reusable UI components
 - Efficient state management
 - Improved application performance
- In the SmartOrder AI project, React.js is used to build:
 - Product pages
 - Cart system
 - Checkout page
 - Admin dashboard
 - AI analytics dashboard
- **Tailwind CSS**

- Tailwind CSS is a **utility-first CSS framework** used to design modern user interfaces quickly.
- Instead of writing large CSS files, Tailwind provides pre-built utility classes to style components directly in HTML or JSX.
- Advantages of Tailwind CSS include:
 - Faster UI development
 - Responsive design
 - Consistent styling
 - Customizable themes
 - Lightweight CSS structure
- In this project, **Tailwind CSS version 4** is used to design:
 - Responsive product cards
 - Navigation bar
 - Dashboard layout
 - Buttons and forms

4.1.2 about HTML

- Hypertext Markup Language (HTML) is the standard markup language for creating web pages and web applications. With Cascading Style Sheets (CSS) and JavaScript it
- HTML (Hypertext Markup Language) is the code that is used **to structure a web page and its content**.
- In the SmartOrder AI project, HTML is used to structure the content of the website including:
 - Product listings
 - Navigation menu
 - Forms for login and checkout
 - Product descriptions
- For example, content could be structured within a set of paragraphs, a list of bulleted points, or using images and data tables.
- forms a triad of cornerstone technologies for the World Wide Web browsers receive HTML documents from a web server or from local storage and render them into multimedia web pages.
- HTML describes the structure of web Page semantically and originally included cues for the appearance of the document.

4.1.3 about CSS

- CSS is the language for describing the presentation of web pages, including colors, layout and fonts.
- In the SmartOrder AI project, CSS works together with **Tailwind CSS** to style the user interface.
- CSS ensures that the website is visually appealing and works properly on different screen sizes such as:
 - Desktop
 - Laptop
 - Tablet
 - Mobile devices
- It allows one to adapt the presentation to different types of devices, such as large screens, small screens, or printers.
- CSS provides these capabilities through a technical model based on a hierarchical scope of effect, the separation of style from content, and a well-defined set of published standards.
- CSS saves a lot of work. It can control the layout of multiple web pages all at once.
- CSS is independent of HTML and can be used with any XML-based mark-up language.

4.1.4 about JavaScript

- JavaScript is a client-side scripting language developed by Brendan
- Each JavaScript can be run on any operating systems and almost all web browsers. The
- Programs in this language are called *scripts*. They can be written right in a web page's HTML.
- In the SmartOrder AI project, JavaScript is used for:
 - Handling user actions
 - Managing cart functionality
 - Sending API requests to the backend
 - Updating UI dynamically
 - Managing application state
- JavaScript is commonly used for creating web pages. It allows us to add dynamic behavior to the webpage and add special effects to the webpage. On websites, it is mainly used for validation purposes.

- And run automatically as the page loads. Scripts are provided and executed as plain text. They don't need special preparation or compilation to run. In this aspect, JavaScript is very different from another language called Java.

4.2 Why Use React.js

- React.js was chosen for the SmartOrder AI project because it provides a powerful framework for building modern web applications.
- The main reasons for using React.js include:

Component-Based Architecture

- React allows developers to create reusable components such as:
 - Navbar
 - Product Card
 - Cart Component
 - Checkout Form

This improves code reusability and maintainability.

Fast Rendering

- React uses a **Virtual DOM** that updates only the parts of the page that change. This improves application performance and reduces unnecessary page reloads.

Easy State Management

- React allows developers to manage application data using states and hooks.
- This is useful for managing:
 - Cart items
 - User login status
 - Product data
 - AI behavior data

Strong Community Support

- React is widely used and supported by a large developer community. This makes development easier and provides access to many tools and libraries.

CHAPTER 5

BACK END OF SYSTEM

5.1 About Back End

- Back End development focuses on the side of the website users can't see. It's what makes a site interactive. The back end can also be referred to as the "server side" of a application.
- As an example, let's say you're running a social media website. You need an accessible place to store all of your users' information.
- In the SmartOrder AI system, the backend performs several important tasks such as:
 - Handling user authentication
 - Managing product data
 - Processing orders
 - Recording payment information
 - Storing AI behavior logs
 - Generating analytics data
- The backend of this system is developed using:
 - **Node.js**
 - **Express.js**
 - **MongoDB**
- A Back End developer will help manage this database and the site contents stored on it.
- This ensures that front end elements on your social media website can continue to function properly as users browse uploaded content and other user profiles.
- The back-end, also called the server side, consists of the server which provides data on request, the application which channels it, and the database which organizes the information.
- For example, when a customer browses shoes on a website, they are interacting with the front end.

5.1.1 About Node.js and MongoDB

- **Node.js**
- Node.js is a **JavaScript runtime environment** that allows developers to run JavaScript code on the server side.
- Node.js is known for its high performance and non-blocking architecture.
- Advantages of Node.js include:
 - Fast execution
 - Asynchronous processing
 - Scalable architecture
 - Large ecosystem of libraries
 - In the SmartOrder AI project, Node.js is used to:
 - Create backend APIs
 - Handle user requests
 - Manage database communication
 - Process AI behavior logs
- **MongoDB**

- MongoDB is a **NoSQL database** used to store application data in a flexible JSON-like format.
- Unlike traditional relational databases, MongoDB stores data in **documents instead of tables**.
- MongoDB is used in this project to store:
 - User information
 - Product data
 - Orders
 - Payment records
 - AI behavior logs
 - Contact messages

5.1.2 How MongoDB Works

- MongoDB stores data using a structure called **collections and documents**.
- Collection
A collection is similar to a table in relational databases.
- Document
A document is a JSON-like data structure that contains key-value pairs.
- MongoDB allows developers to store structured and unstructured data efficiently.
- In the SmartOrder AI system, MongoDB collections include:
 - Users Collection
 - Products Collection
 - Orders Collection
 - Payments Collection
 - AI Logs Collection

5.1.3 MongoDB Features

- MongoDB provides several features that make it suitable for modern web applications.
 - Flexible Schema
 - MongoDB allows developers to store data without strict table structures.
 - High Performance
 - MongoDB supports fast data retrieval and indexing.
 - Scalability
 - MongoDB can handle large datasets and high traffic applications.
 - JSON-Based Data
 - MongoDB uses a document format similar to JSON, which integrates well with JavaScript applications.
 - Easy Integration
 - MongoDB works easily with Node.js applications.

5.2 Why Use MongoDB and Node.js

- MongoDB and Node.js are commonly used together to build modern web applications.
- The reasons for using them in the SmartOrder AI project include:
 - Full JavaScript Stack
 - Both frontend and backend use JavaScript, which simplifies development.
- Fast Data Processing
 - Node.js handles asynchronous requests efficiently, making it suitable for high-performance applications.
- Flexible Data Storage
 - MongoDB allows developers to store complex data structures easily.
- Scalability
 - The combination of Node.js and MongoDB allows the system to scale easily when the number of users increases.

CHAPTER 6

SYSTEM DESIGN

6.1 Data Flow Diagram

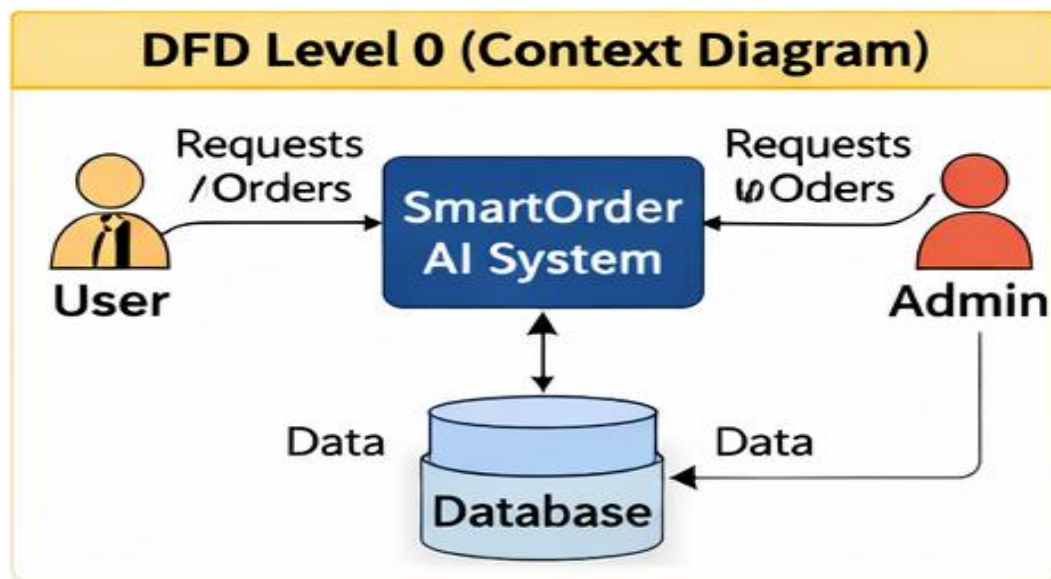


Fig 6.1 Data Flow Diagram LEVEL 0

DFD LEVEL 1: ADMIN

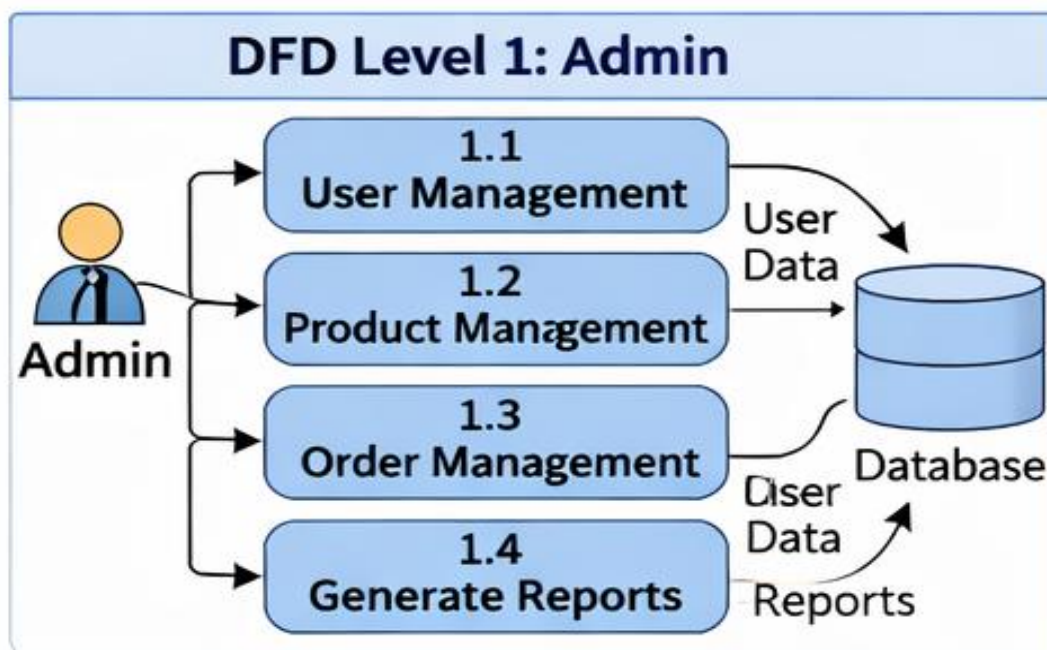


Fig 6.1 Data Flow Diagram LEVEL 1-Admin

DFD LEVEL 1: User

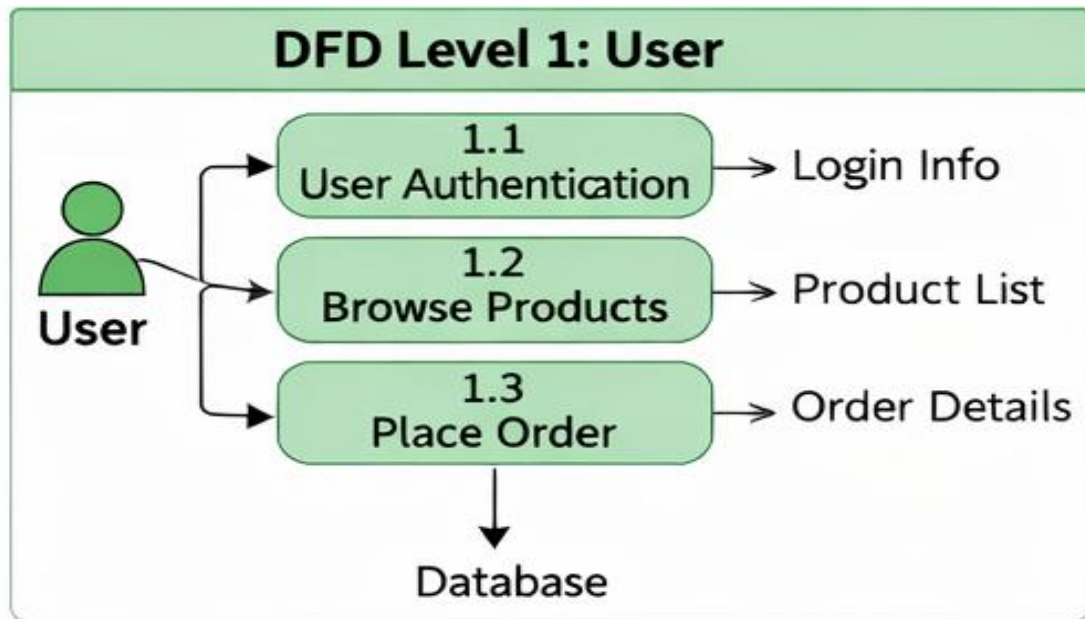


Fig 6.1 Data Flow Diagram LEVEL 1-User

6.2 Use Case Diagram

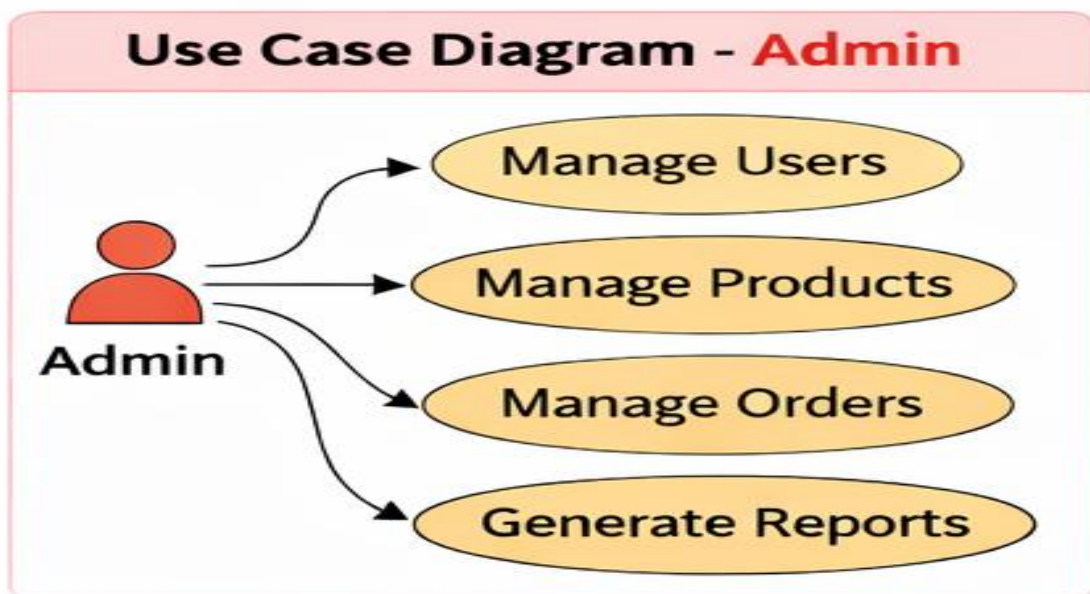


Fig 6.2 Use Case Diagram Of Admin

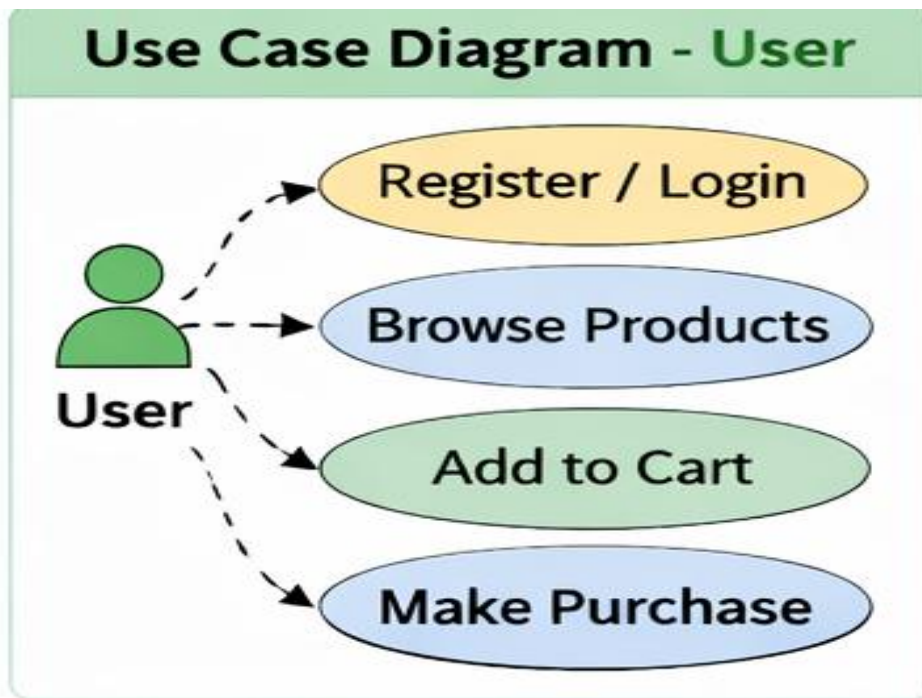


Fig 6.2 Use Case Diagram Of User

6.3 Activity Diagram

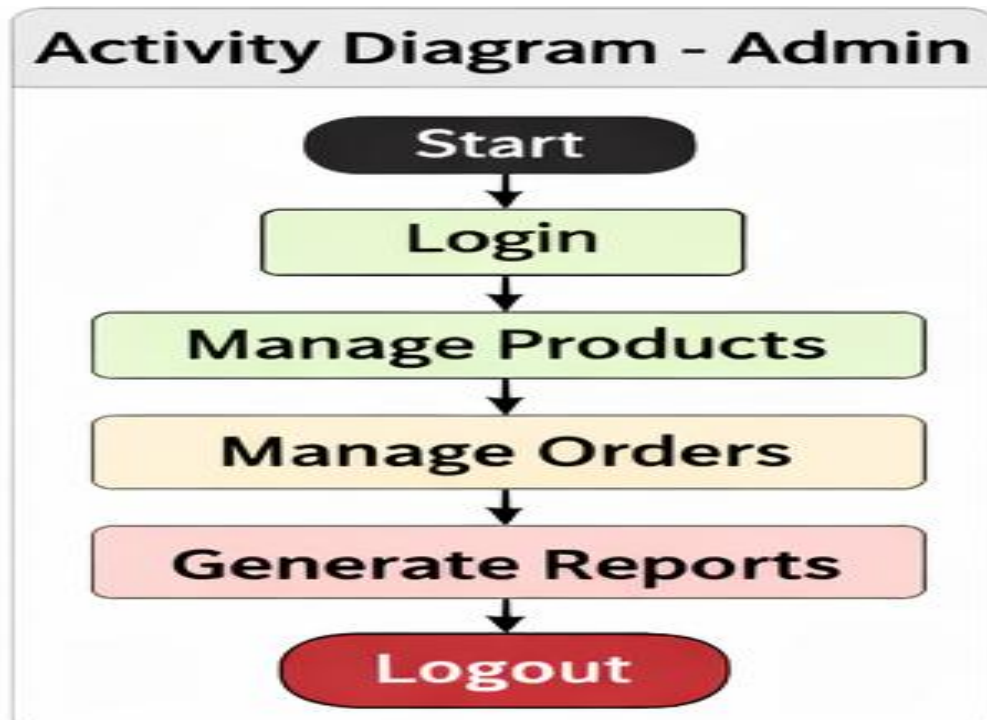


Fig 6.3 Activity diagram of Admin

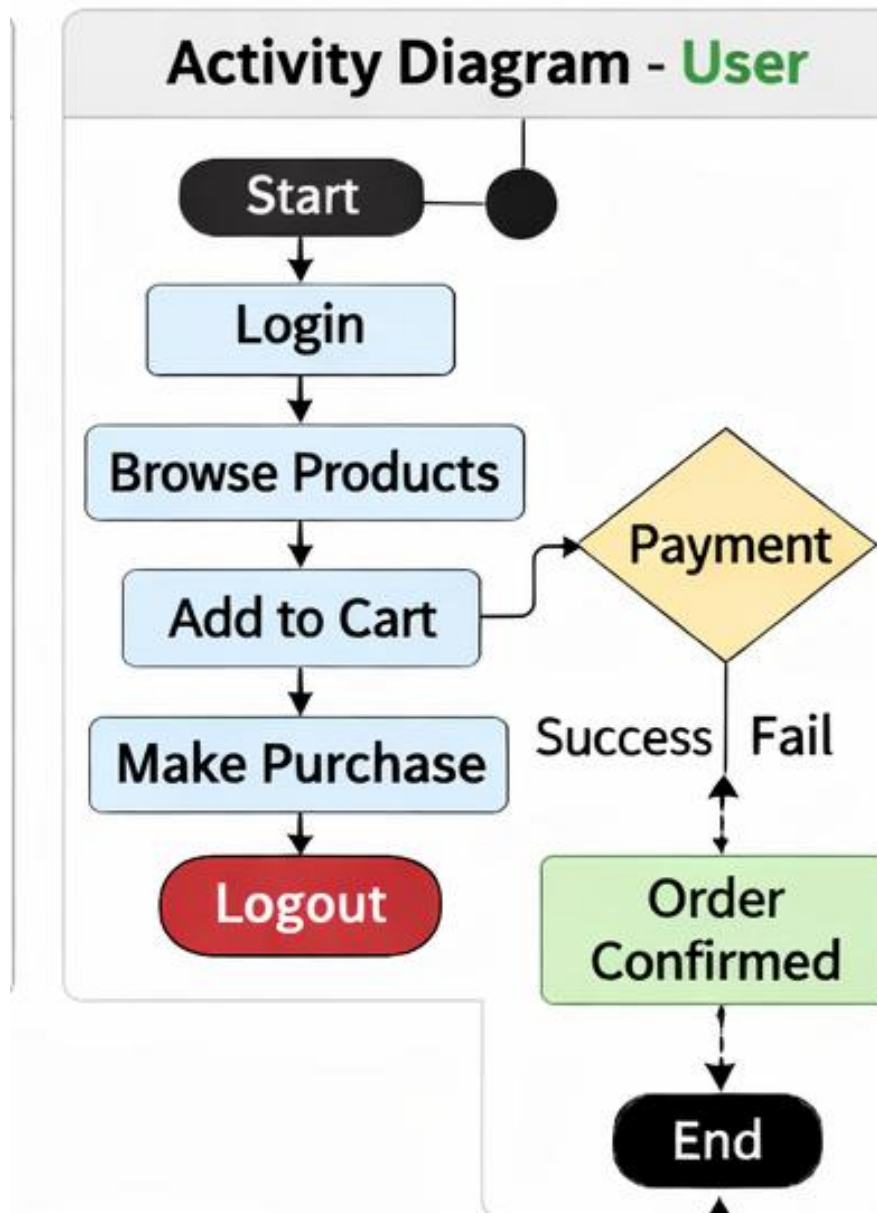


Fig 6.3 Activity Diagram Of User

6.4 E-R Diagram

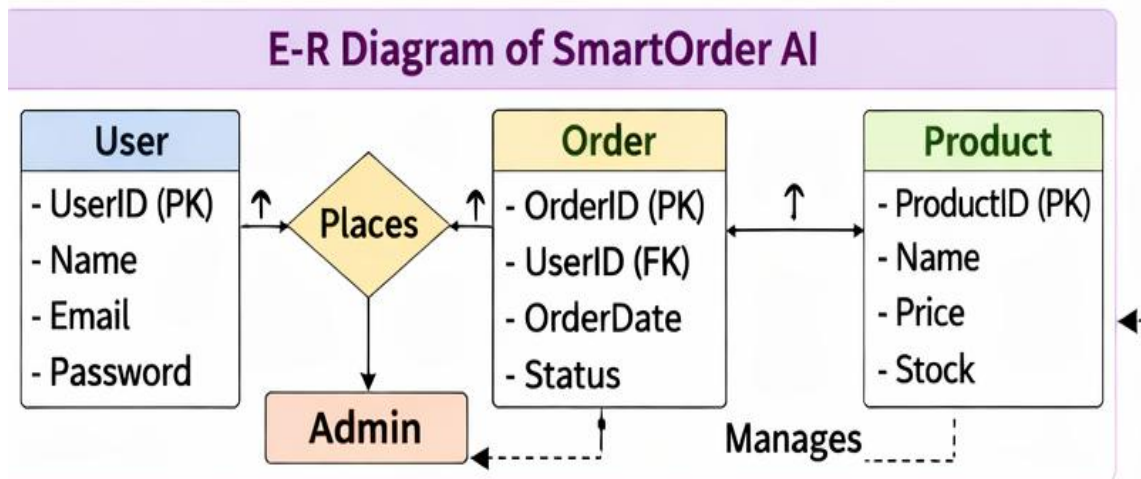


Fig 6.4 Er diagram

CHAPTER 7

DATA DICTIONARY

7.1 Introduction

- Database design is the organization of data according to a database model. The designer determines what data must be stored and how the data elements interrelate.
- A **Data Dictionary** is a detailed description of all the data elements used in the system. It defines the structure of the database tables, attributes, data types, and their purpose in the system.
- The data dictionary helps developers, administrators, and users understand how data is stored and used within the system.
- In the **SmartOrder AI project**, MongoDB is used as the database to store application data. The system stores information related to users, products, orders, payments, and AI behavior logs.
- Each table or collection stores specific information required for the functioning of the system.
- The data dictionary provides details about:
 - Table name
 - Field name
 - Data type
 - Description of the field
 - This helps maintain consistency and clarity in the database design.

7.2 List of Tables

7.2.1 Login Table

Table No: -1

Table Name: -login

Description: -The **Login table** stores user authentication information that allows users and administrators to access the system.

Primary Key: -id

Field Name	Data Type	Description
login_id	ObjectId	Unique login ID
email	string	User email address
Password	string	User password

role	String	User role (Admin/User)
created_at	Date	Account creation date

7.2.2 State Table

Table No: - 2

Table Name: - State

Description: -The **State table** stores information about different states used in the system for address details.

Primary Key: -id

Field Name	Data Type	Description
state_id	ObjectId	Unique state ID
state_name	string	Name of the state
created_at	Date	Date when state record was created

Example:

- Gujarat
- Maharashtra
- Rajasthan

7.2.3 City Table

Table No: - 3

Table Name: - city

Description: - The **City table** stores information about cities belonging to different states.

Primary Key: -id

Field Name	Data Type	Description
city_id	ObjectId	Unique city ID
city_name	string	Name of the city
State_id	ObjectId	Reference to state
created_at	Date	Date when city was added

Example:

- Ahmedabad
- Surat
- Mumbai

7.2.4 Registration Table

Table No: - 4

Table Name: - registration

Description: -The **Registration table** stores detailed information about registered users of the system.

Primary Key: -id

Field Name	Data Type	Description
user_id	ObjectId	Unique user ID
first_name	string	User first name
last_name	String	User last name
email	String	Email address

password	String	User password
phone	String	Contact number
address	String	User address
City_id	ObjectId	Reference to city
State_id	ObjectId	Reference to state
created_at	Date	Registration date

7.2.5 T-Shirt Booking Table

Table No: -5

Table Name: -T-Shirt Booking Table

Description: -The **T-Shirt Booking table** stores information about customer orders for T-shirts.

Primary Key: -id

Field Name	Data Type	Description
booking_id	ObjectId	Unique booking ID
user_id	ObjectId	Reference to user
product_id	ObjectId	Reference to product
t_shirt_name	String	Name of T-shirt

color	String	Color of T-shirt
size	String	T-shirt size
price	Numbet	Product price
quantity	Number	Number of items
booking_date	Date	Order date
order_status	String	Order status

7.2.6 SmartOrder AI Table

Table No: - 6

Table Name: - SmartOrder AI Table

Description: - The **SmartOrder AI table** stores behavioral data generated by the AI system during user interactions.

This table is used to analyze whether a purchase is **Rational or Impulse**.

Primary Key: -id

Field Name	Data Type	Description
ai_log_id	ObjectId	Unique AI log ID
user_id	ObjectId	Reference to user
product_id	ObjectId	product viewed
view_time	Number	Time spent on product page

interaction_score	Number	User interaction score
decision_score	Numbet	AI decision score
purchase_type	Number	Rational / Impulse
created_at	Date	Log creation time

This table is used to generate **AI analytics dashboards**.

7.2.7 Inquiry Table

Table No: -7

Table Name: - inquiry

Description: -The **Inquiry table** stores messages sent by users through the contact form.

Primary Key: -id

Field Name	Data Type	Description
inquiry_id	ObjectId	Unique inquiry ID
name	String	User name
email	String	User email
subject	String	Message subject
message	string	User message
created_at	Date	Inquire submission date

7.2.8 Feedback Table

Table No: -8

Table Name: - Feedback

Description: -The **Feedback table** stores customer feedback about the system and products.

Primary Key: -id

Field Name	Data Type	Description
feedback_id	ObjectId	Unique feedback ID
user_id	ObjectId	Reference to user
rating	Number	Product rating
comment	String	Feedback message
created_at	Date	Feedback submission date

This data helps administrators understand user satisfaction and improve the platform.

CHAPTER 8

TESTING

8.1 Testing Plan

- A Test Plan is a detailed document that describes the test strategy, objectives, schedule, estimation, deliverables, and resources required to perform testing for a software product.
- Testing is an important phase of software development that ensures the system works correctly and meets user requirements.
- The main purpose of testing is to identify errors, bugs, or issues in the system before the final deployment. It helps verify that all features of the application work properly and provide the expected output.
- In the SmartOrder AI system, testing was performed on both the frontend and backend components to ensure proper functionality.
- Testing was carried out on the following modules:
 - User Login System
 - Product Browsing
 - Cart Management
 - Checkout Process
 - Order Processing
 - AI Behavior Tracking
 - Admin Dashboard
- The testing process ensured that the system performs efficiently, securely, and reliably.
- Test Plan helps us determine the effort needed to validate the quality of the application under test.
- The test plan serves as a blueprint to conduct software testing activities as a defined process, which is minutely monitored and controlled by the test manager.
- Test Plan is A document describing the scope, approach, resources, and schedule of intended test activities

8.2 Testing Strategies

- A Test Strategy is a plan for defining an approach to the Software Testing Life Cycle (STLC).
- It helps testers get a clear picture of the project at any instance.
- Testing strategies define the approach used to test the system effectively.
- In the SmartOrder AI project, the following testing strategies were used:

➤ Unit Testing

- **Unit testing focuses** on testing individual components or modules of the system.

Examples:

- Login module testing

- Product display module testing
 - Cart functionality testing
- The possibility of missing any test activity is very low when there is a proper test strategy in place.
- Test Strategy Document is a well-described document in software testing which clearly defines the exact software testing approach and testing objectives of the software application.
- Testing strategy plan should be communicated with the entire team so that the team will be consistent on approach and responsibilities.

8.3 Integration Testing

- Integration testing verifies that different modules of the system work correctly when combined together.

Examples:

- Frontend interacting with backend APIs
- Backend interacting with MongoDB database
- Checkout process connecting with order database

8.4 System Testing

- System testing evaluates the entire application to ensure that all features work correctly together.

Examples:

- Complete user purchase process
- Admin dashboard functionality
- AI behavior analysis results

8.5 User Interface Testing

- UI testing ensures that the user interface is responsive, accessible, and easy to use.

This includes testing:

- Navigation menus
- Buttons and forms
- Product pages
- Checkout interface

8.6 Testing Method

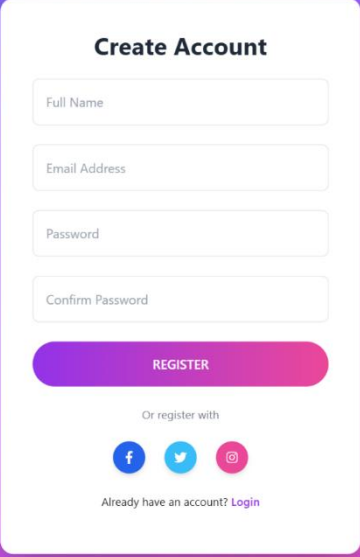
- Software testing methodologies are the various strategies or approaches used to test an application to ensure it behaves and looks as expected.
- The testing method used for this project is **Black Box Testing**.
- Black Box Testing focuses on testing the system functionality without examining the internal code structure.

- The tester provides inputs and verifies whether the system produces the correct outputs.
- Examples in SmartOrder AI:
 - Input
User enters login credentials
 - Expected Output
User should successfully login if credentials are correct.
 - Input
User adds product to cart
 - Expected Output
Product should appear in cart with correct price and quantity.
- Black Box Testing helps ensure that the system behaves correctly from the user's perspective.
- Software Testing Methodology is defined as strategies and testing types used to certify that the Application Under Test meets client expectations.
Each testing methodology has a defined test objective, test strategy, and deliverables.

8.4 Test Case

- A test case is nothing but a series of step executed on a product, using a predefined set of input data, expected to produce a pre-defined set of outputs, in a given environment.
- It describes “how” to implement those test cases.
- A **Test Case** is a set of conditions used to verify whether a particular feature of the system works correctly.
- Below are sample test cases for the SmartOrder AI system.
- Test case specifications are useful as it enlists the specification details of the items.

Test Case 1 – User Login: -



Create Account

Full Name

Email Address

Password

Confirm Password

REGISTER

Or register with

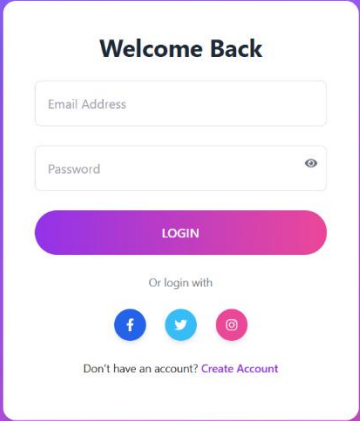




Already have an account? [Login](#)

Test Case ID	TC01
Module	Register System
Input	Valid Name,Email and Password
Expected Result	User should Register successfully
Actual Result	Register successful
Status	Pass

Test Case 2 – Invalid Login



Welcome Back

Email Address

Password

LOGIN

Or login with





Don't have an account? [Create Account](#)

Test Case ID	TC02
Module	Login System
Input	valid Email or Password
Expected Result	Login successful

Actual Result	Login message displayed
Status	Pass

Test Case 3 – Add Product to Cart

SmartOrder AI

AI ASSISTED DECISION INTELLIGENCE

SHOP

Blue

Black

White

About

Contact

AI

CART

Journal

Checkout

CONTROL HUB

LOGOUT

shirt

₹230

In Stock

Experience effortless style. Crafted from breathable, soft premium cotton, ideal for everyday comfort and styling.

Color

Size

S

M

L

XL

XXL

Quantity

-

1

+

ADD TO CART

Get connected with us on social networks

SmartOrder AI

SmartOrder AI is an intelligent e-commerce platform that uses AI-driven decision intelligence to reduce impulse buying and improve customer experience.

Products

Blue Collection

Black Collection

White Collection

Shopping Cart

Useful Links

About Us

Contact Us

Admin Login

Dashboard

Contact

WhatsApp: 09494111111

support@smartorderai.com

+91 9896462100

AI Decision Support System

© 2025 SmartOrder AI | All Rights Reserved

Shopping Cart

shirt

₹230

Color: Black

Size: M

Quantity: 1

Subtotal (1 items)

₹230.00

GST (18%)

₹41.40

Shipping

Free

Estimated Total

₹271.40

PROCESS TO INTELLIGENT RECOMMENDATION

PROCESS TO CHECKOUT

Get connected with us on social networks

SmartOrder AI

SmartOrder AI is an intelligent e-commerce platform that uses AI-driven decision intelligence to reduce impulse buying and improve customer experience.

Products

Blue Collection

Black Collection

White Collection

Shopping Cart

Useful Links

About Us

Contact Us

Admin Login

Dashboard

Contact

WhatsApp: 09494111111

support@smartorderai.com

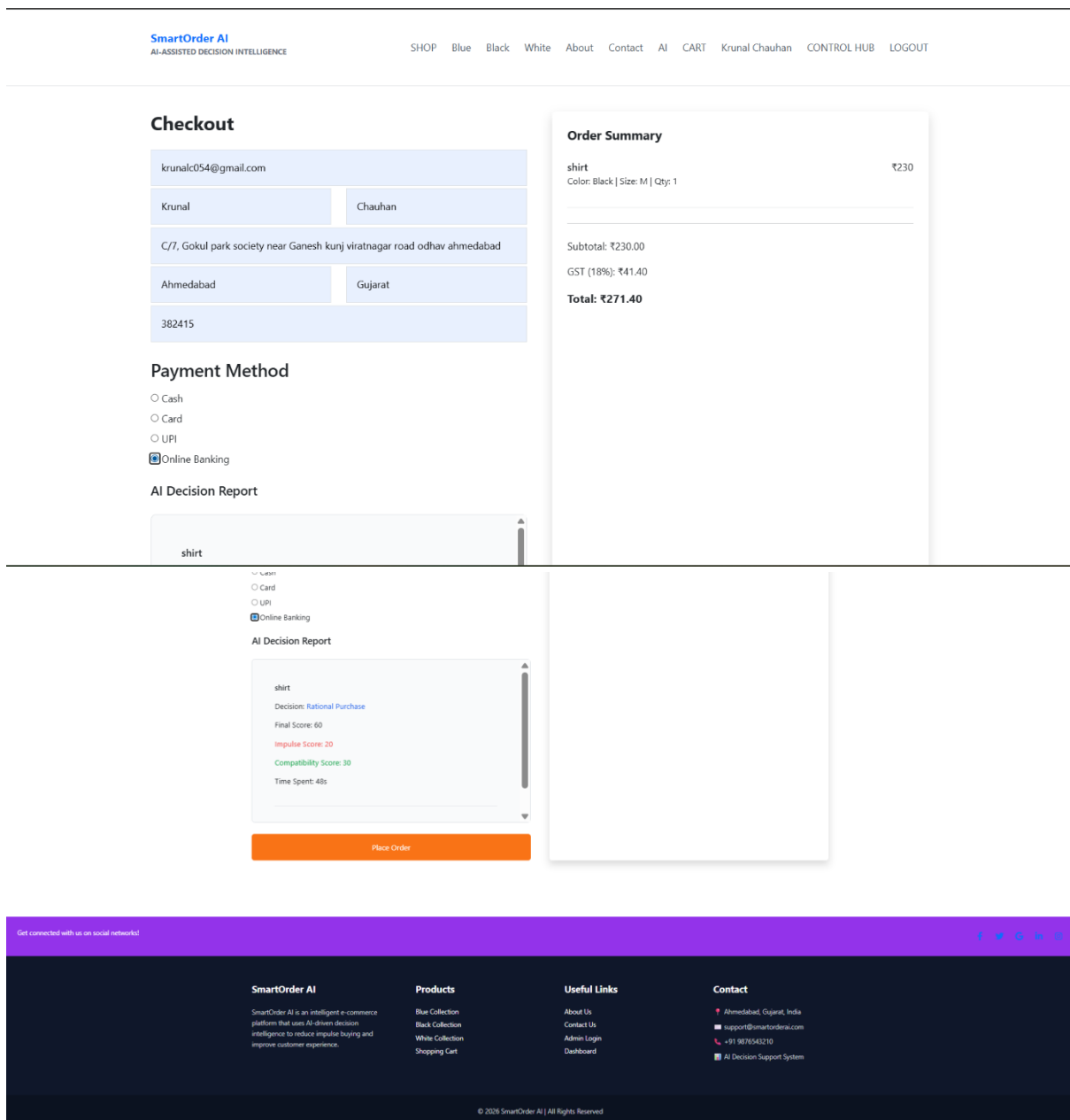
+91 9896462100

AI Decision Support System

© 2025 SmartOrder AI | All Rights Reserved

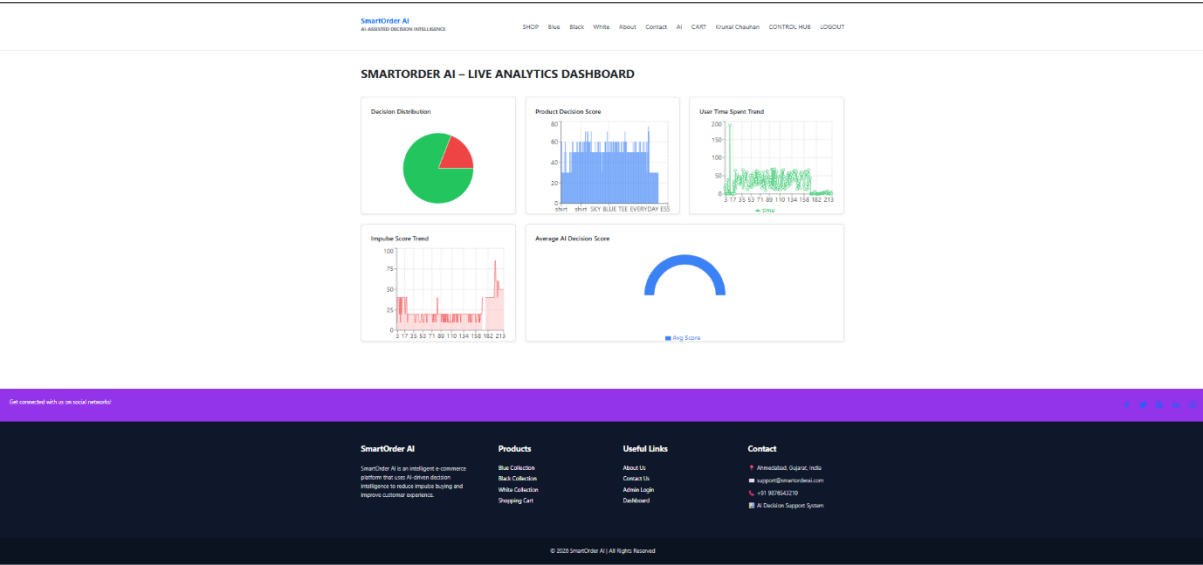
Test Case ID	TC03
Module	Cart System
Input	Click "Add to Cart" button
Expected Result	Product added to cart
Actual Result	Product added successfully
Status	Pass

Test Case 4 – Checkout Process



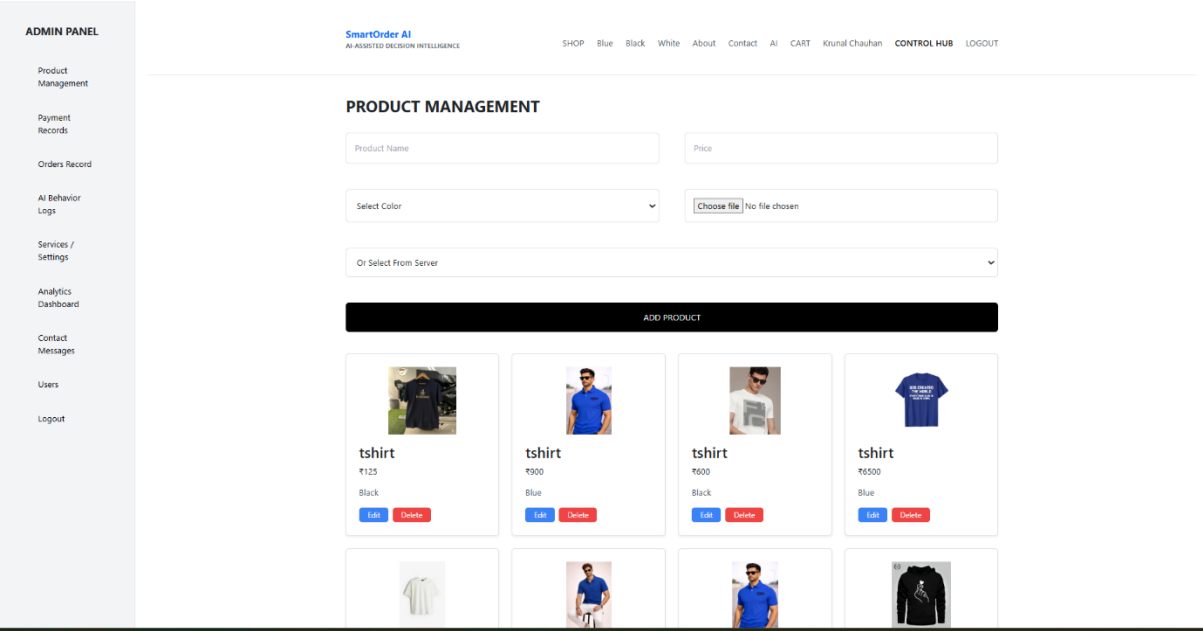
Test Case ID	TC04
Module	Checkout System
Input	User fills checkout form and confirms order
Expected Result	Order should be stored in database
Actual Result	Order saved successfully
Status	Pass

Test Case 5 – AI Behavior Tracking



Test Case ID	TC05
Module	AI Behavior System
Input	User views product and adds to cart
Expected Result	Behavior data stored in AI logs
Actual Result	Data recorded successfully
Status	Pass

Test Case 6 – Admin Product Management



ADMIN PANEL

Product Management

Payment Records

Orders Record

AI Behavior Logs

Services / Settings

Analytics Dashboard

Contact Messages

Users

Logout

Payment Records

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: shirt	
Amount: ₹271.4	Status: Success
Date: 16/03/2026, 13:02:22	

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: SKY BLUE TEE - 2	
Amount: ₹901.52	Status: Success
Date: 14/03/2026, 12:13:42	

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: SKY BLUE TEE - 14	
Amount: ₹1113.92	Status: Success
Date: 11/03/2026, 13:18:51	

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: SKY BLUE TEE - 15	
Amount: ₹1131.62	Status: Success
Date: 11/03/2026, 13:17:44	

Name: Krunal Chauhan	Email: krunal054@gmail.com
-----------------------------	-----------------------------------

ADMIN PANEL

Product Management

Payment Records

Orders Record

AI Behavior Logs

Services / Settings

Analytics Dashboard

Contact Messages

Users

Logout

Orders Record

Order ID	Customer	Email	Products	Amount	Payment	Status	Date
ORD-1773646342115	Krunal Chauhan	krunal054@gmail.com	shirt	₹271.4	Online Banking	Completed	16/03/2026, 13:02:22
ORD-1773470622478	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 2	₹901.52	Cash	Completed	14/03/2026, 12:13:42
ORD-1773215331742	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 14	₹1113.92	Cash	Completed	11/03/2026, 13:18:51
ORD-1773215264219	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 15	₹1131.62	Cash	Completed	11/03/2026, 13:17:44
ORD-1773215203817	Krunal Chauhan	krunal054@gmail.com	CLOUD WHITE TEE - 7	₹952.26	Card	Completed	11/03/2026, 13:16:43
ORD-1773214768959	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 16	₹1149.32	Card	Completed	11/03/2026, 13:09:28
ORD-1773214713048	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 17, CLOUD WHITE TEE - 10	₹3350.02	Cash	Completed	11/03/2026, 13:08:33
ORD-1773214670042	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 14, SKY BLUE TEE - 11	₹2192.44	Online Banking	Completed	11/03/2026, 13:07:50
ORD-1773214644813	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 14	₹1131.62	Cash	Completed	11/03/2026, 13:07:24
ORD-1773214606434	Krunal Chauhan	krunal054@gmail.com	CLOUD WHITE TEE - 3, PREMIUM BLACK TEE - 7	₹1833.72	Cash	Completed	11/03/2026, 13:06:46
ORD-1773214570155	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 10	₹1043.12	Cash	Completed	11/03/2026, 13:06:10
ORD-1773214396066	Krunal Chauhan	krunal054@gmail.com	T shirt	₹944	Card	Completed	11/03/2026, 13:03:10
ORD-1773214302359	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 2	₹648.42	Cash	Completed	11/03/2026, 13:01:42
ORD-1773214276236	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 5	₹1909.24	Cash	Completed	11/03/2026, 13:01:16
ORD-1773214214760	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 4	₹936.92	Cash	Completed	11/03/2026, 13:00:14
ORD-1773214189916	Krunal Chauhan	krunal054@gmail.com	CLOUD WHITE TEE - 3	₹867.3	Card	Completed	11/03/2026, 12:59:49
ORD-1773214142691	Krunal Chauhan	krunal054@gmail.com	tshirt	₹2360	Online Banking	Completed	11/03/2026, 12:59:02
ORD-1773213924668	Krunal Chauhan	krunal054@gmail.com	tshirt	₹1062	Cash	Completed	11/03/2026, 12:55:24
ORD-1773213905584	Krunal Chauhan	krunal054@gmail.com	tshirt	₹590	Cash	Completed	11/03/2026, 12:55:05
ORD-1773213853202	Krunal Chauhan	krunal054@gmail.com	shirt	₹271.4	Cash	Completed	11/03/2026, 12:54:13
ORD-1773127750108	Krunal Chauhan	krunal054@gmail.com	tshirt	₹4248	Online Banking	Completed	10/03/2026, 12:59:10
ORD-1773127355194	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 4, SKY BLUE TEE - 3	₹1814.84	Cash	Completed	10/03/2026, 12:52:35
ORD-1773127193546	Krunal Chauhan	krunal054@gmail.com	tshirt	₹1180	Cash	Completed	10/03/2026, 12:49:53

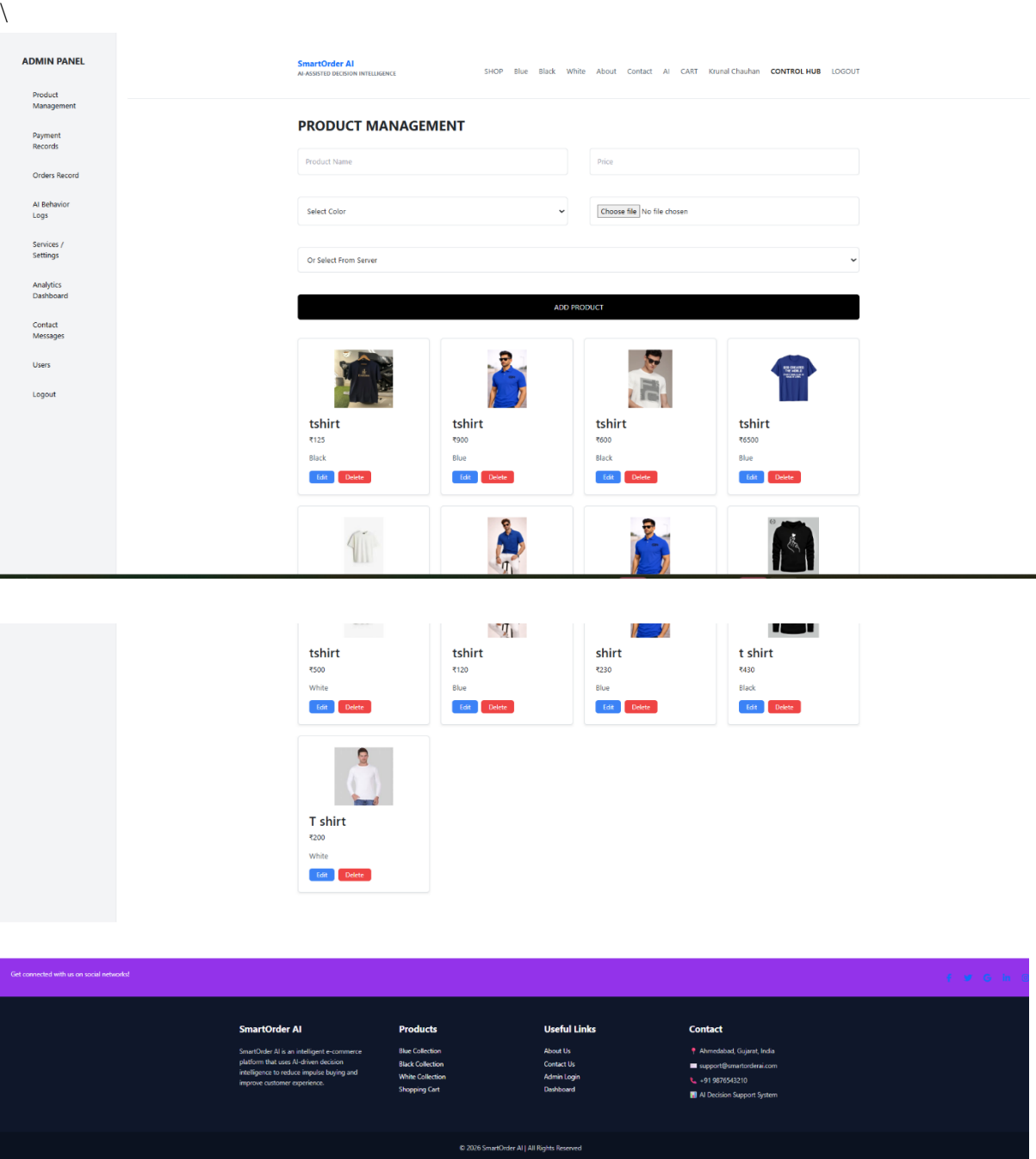
Test Case ID	TC06
Module	Admin Dashboard
Input	Admin adds new product
Expected Result	Product should appear in product list
Actual Result	Product added successfully
Status	Pass

CHAPTER 9

SNAPSHOT OF WEBSITE

9.1 Admin Site

❖ PRODUCT MANAGEMENT: -



❖ PAYMENTS RECORDS: -

ADMIN PANEL

Product Management

Payment Records

Orders Record

AI Behavior Logs

Services / Settings

Analytics Dashboard

Contact Messages

Users

Logout

Payment Records

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: shirt	
Amount: ₹271.4	Status: Success
Date: 16/03/2026, 13:02:22	

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: SKY BLUE TEE - 2	
Amount: ₹901.52	Status: Success
Date: 14/03/2026, 12:13:42	

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: SKY BLUE TEE - 14	
Amount: ₹1113.92	Status: Success
Date: 11/03/2026, 13:18:51	

Name: Krunal Chauhan	Email: krunal054@gmail.com
Product: SKY BLUE TEE - 15	
Amount: ₹1131.62	Status: Success
Date: 11/03/2026, 13:17:44	

Name: Krunal Chauhan	Email: krunal054@gmail.com
-----------------------------	-----------------------------------

❖ ORDERS RECORD: -

ADMIN PANEL

Product Management

Payment Records

Orders Record

AI Behavior Logs

Services / Settings

Analytics Dashboard

Contact Messages

Users

Logout

Order ID	Customer	Email	Products	Amount	Payment	Status	Date
ORD-1773646342115	Krunal Chauhan	krunal054@gmail.com	shirt	₹271.4	Online Banking	Completed	16/03/2026, 13:02:22
ORD-1773470622478	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 2	₹901.52	Cash	Completed	14/03/2026, 12:13:42
ORD-1773215331742	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 14	₹1113.92	Cash	Completed	11/03/2026, 13:18:51
ORD-1773215264219	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 15	₹1131.62	Cash	Completed	11/03/2026, 13:17:44
ORD-1773215203817	Krunal Chauhan	krunal054@gmail.com	CLOUD WHITE TEE - 7	₹952.26	Card	Completed	11/03/2026, 13:16:43
ORD-1773214768959	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 16	₹1149.32	Card	Completed	11/03/2026, 13:09:28
ORD-1773214713048	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 17, CLOUD WHITE TEE - 10	₹3350.02	Cash	Completed	11/03/2026, 13:08:33
ORD-1773214670042	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 14, SKY BLUE TEE - 11	₹2192.44	Online Banking	Completed	11/03/2026, 13:07:50
ORD-1773214644813	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 14	₹1131.62	Cash	Completed	11/03/2026, 13:07:24
ORD-1773214606434	Krunal Chauhan	krunal054@gmail.com	CLOUD WHITE TEE - 3, PREMIUM BLACK TEE - 7	₹1833.72	Cash	Completed	11/03/2026, 13:06:46
ORD-1773214570155	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 10	₹1043.12	Cash	Completed	11/03/2026, 13:06:10
ORD-1773214396066	Krunal Chauhan	krunal054@gmail.com	T shirt	₹944	Card	Completed	11/03/2026, 13:03:10
ORD-1773214302359	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 2	₹848.42	Cash	Completed	11/03/2026, 13:01:42
ORD-1773214276236	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 5	₹1909.24	Cash	Completed	11/03/2026, 13:01:16
ORD-1773214214760	Krunal Chauhan	krunal054@gmail.com	SKY BLUE TEE - 4	₹936.92	Cash	Completed	11/03/2026, 13:00:14
ORD-1773214189916	Krunal Chauhan	krunal054@gmail.com	CLOUD WHITE TEE - 3	₹867.3	Card	Completed	11/03/2026, 12:59:49
ORD-1773214142691	Krunal Chauhan	krunal054@gmail.com	tshirt	₹2360	Online Banking	Completed	11/03/2026, 12:59:02
ORD-1773213924668	Krunal Chauhan	krunal054@gmail.com	tshirt	₹1062	Cash	Completed	11/03/2026, 12:55:24
ORD-1773213905384	Krunal Chauhan	krunal054@gmail.com	tshirt	₹590	Cash	Completed	11/03/2026, 12:55:05
ORD-1773213853202	Krunal Chauhan	krunal054@gmail.com	shirt	₹271.4	Cash	Completed	11/03/2026, 12:54:13
ORD-17732127750108	Krunal Chauhan	krunal054@gmail.com	tshirt	₹4248	Online Banking	Completed	10/03/2026, 12:59:10
ORD-17732127355194	Krunal Chauhan	krunal054@gmail.com	PREMIUM BLACK TEE - 4, SKY BLUE TEE - 3	₹1814.84	Cash	Completed	10/03/2026, 12:52:35
ORD-17732127193546	Krunal Chauhan	krunal054@gmail.com	tshirt	₹1180	Cash	Completed	10/03/2026, 12:49:53

❖ AI BEHAVIOR LOGS(LIVE):-

ADMIN PANEL

Product Management

Payment Records

Orders Record

AI Behavior Logs

Services / Settings

Analytics Dashboard

Contact Messages

Users

Logout

AI Behavior Logs (Live)

Total Sessions

213

Rational Purchases

157

Impulse Purchases

37

EVERYDAY ESSENTIAL - CLOUD WHITE

Impulse Score: 50

Time Spent: 1s

Size: M

Action: add_to_cart

Date: 02/03/2026, 14:46:38

Compatibility: 15

Color: Black

Quantity: 1

CLASSIC CREWNECK - SKY BLUE

Impulse Score: 50

Time Spent: 3s

Size: M

Action: add_to_cart

Date: 02/03/2026, 14:47:18

Compatibility: 15

Color: Black

Quantity: 1

❖ SERVICE & SETTINGS: -

ADMIN PANEL

Product Management

Payment Records

Orders Record

AI Behavior Logs

Services / Settings

Analytics Dashboard

Contact Messages

Users

Logout

Services & Settings

Total Orders

42

Total Revenue

₹61913.42

System Status

Live & Running

AI Configuration

AI Decision Threshold (%)

Current Threshold: 40%

Website Maintenance Mode

☐ Enable Maintenance Mode

Admin Account Settings

Admin Name

Admin Email

New Password

Save All Settings

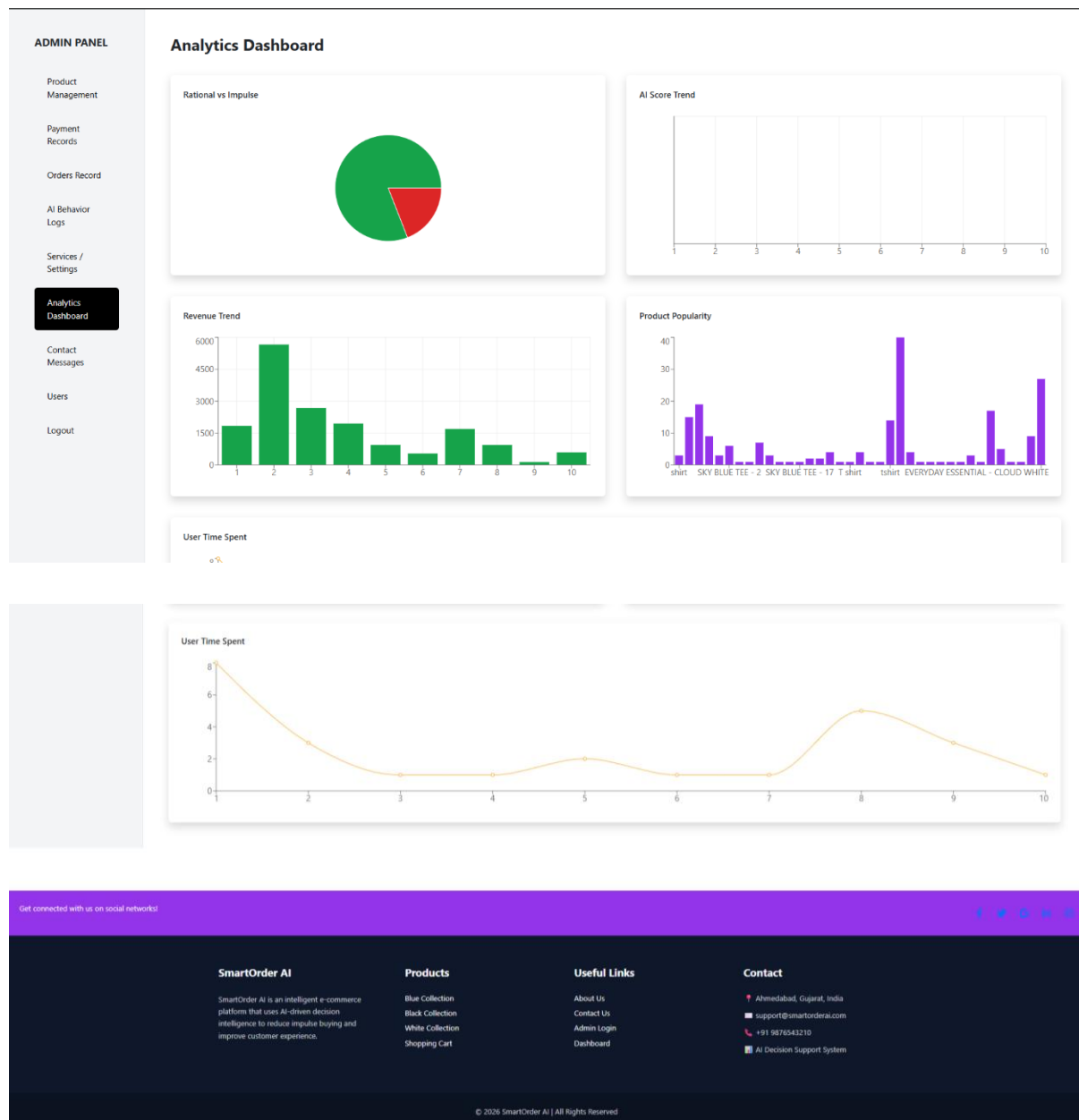
System Information

Version: SmartOrder AI v1.0

Server: Running on Port 8000

Database: MongoDB Connected

❖ ANALYTICS DASHBOARD



❖ REGISTER USERS: -

ADMIN PANEL

Product Management

Payment Records

Orders Record

AI Behavior Logs

Services / Settings

Analytics Dashboard

Contact Messages

Users

Logout

Registered Users

#	Name	Email
1	Test User	test123@gmail.com
2	Krunal Chauhan	krunalc054@gmail.com
3	Krunal Chauhan	krunalc05@gmail.com
4	Krunal Chauhan	krunalc045@gmail.com
5	Krunal Chauhan	krunal@gmail.com
6	Krunal Chauhan	aa@gmail.com
7	Krunal Chauhan	bb@gmail.com
8	Test User	test12@gmail.com
9	Krunal Chauhan	as@gmail.com
10	Krunal Chauhan	viren@gmail.com
11	Krunal Chauhan	Hi@gmail.com
12	admin	admin@gmail.com
13	het	het@gmail.com
14	Krunal Chauhan	kc@gmail.com

❖ CONTACT MESSAGES: -

ADMIN PANEL		Contact Messages	
Product Management		Name: Krunal Chauhan Phone: 5632563210 Message: hello hi 04/03/2026, 11:12:23	Email: krunal@gmail.com
Payment Records			
Orders Record			
AI Behavior Logs		Name: Krunal Chauhan Phone: 09316061087 Message: user 03/03/2026, 18:22:54	Email: krunalc054@gmail.com
Services / Settings			
Analytics Dashboard			
Contact Messages		Name: Krunal Chauhan Phone: 09316061087 Message: hi 03/03/2026, 15:08:11	Email: krunalc054@gmail.com
Users			
Logout			

❖ **View All Users: -**
 This page shows all users.

ePASS

Dashboard

Pass Types

Passes Request

All Users

All Inquiries

Running Passes

Logout

ALL USERS

ID	Name	Email	Contact	State	City	Gender
1	Alkesh Alkesh	kabaalkesh293@gmail.com	0901664748	Gujarat	Ahmedabad	male
2	Hitesh Hitesh	hiteshkachhela@gmail.com	9016395600			
3	Praxware Praxware	a@yahoo.in	0901664748	Gujarat	Ahmedabad	female
4	Umesh Umesh	umesh@gmail.com	9725883312			
5	rinku h rinku h	rinkuparmar2097@gmail.com	9408424199	Gujarat	Ahmedabad	female
6	Alkesh Alkesh	sakshi@gmail.com	0901664748	Gujarat	Surat	male
7	anamika anamika	anamika@gmail.com	6353410733			

© 2020 Praxware Technologies. All rights reserved

❖ View All Inquiry: -

This page show all inquiry.

ePASS

Dashboard

Pass Types

Passes Request

All Users

All Inquiries

Running Passes

Logout

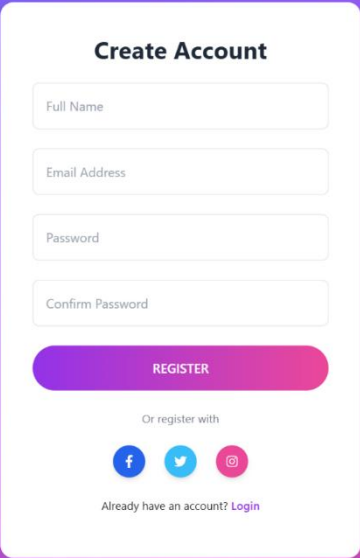
ALL INQUIRIES

ID	Name	Email	Contact	Subject	Message	Date
1	Alkesh	kabaalkesh293@gmail.com	9016647480	Testing	This is for Testing	2019-03-15
2	Hitesh	hitesh@gmail.com	9016395600	Testing Purpose	This is Description	2019-03-18

© 2020 Praxware Technologies. All rights reserved

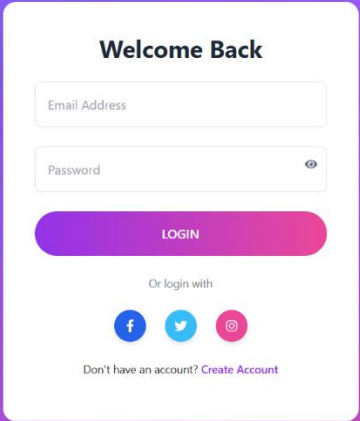
9.2 Users Site

❖ REGISTER PAGE: -



The registration form is titled "Create Account" and is centered on a purple-to-pink gradient background. It contains four input fields: "Full Name", "Email Address", "Password", and "Confirm Password". Below these fields is a prominent "REGISTER" button with a pink-to-purple gradient. Underneath the button, the text "Or register with" is followed by three social media icons: Facebook, Twitter, and Instagram. At the bottom, a link reads "Already have an account? [Login](#)".

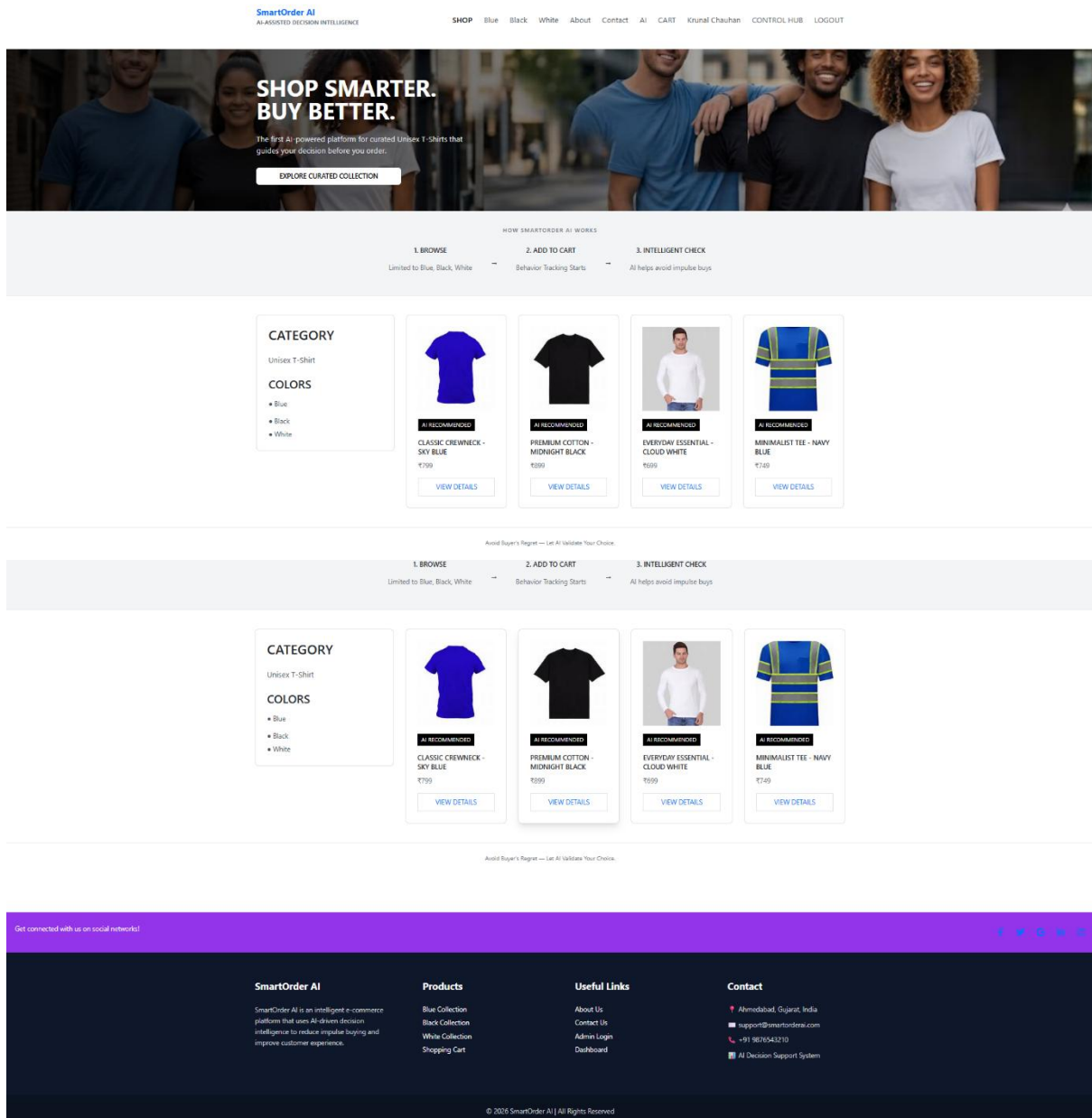
❖ LOGIN PAGE: -



The login form is titled "Welcome Back" and is centered on the same purple-to-pink gradient background. It features two input fields: "Email Address" and "Password", with an eye icon for password visibility. A "LOGIN" button with a pink-to-purple gradient is positioned below the fields. Underneath the button, the text "Or login with" is followed by three social media icons: Facebook, Twitter, and Instagram. At the bottom, a link reads "Don't have an account? [Create Account](#)".

❖ HOME PAGE: -

This page show services.



❖ **PRODUCT PAGE: -**

Blue T-Shirts Collection



AI RECOMMENDED

SKY BLUE TEE - 1

₹749

VIEW DETAILS



AI RECOMMENDED

SKY BLUE TEE - 2

₹764

VIEW DETAILS



AI RECOMMENDED

SKY BLUE TEE - 3

₹779

VIEW DETAILS



AI RECOMMENDED

SKY BLUE TEE - 4

₹794

VIEW DETAILS



AI RECOMMENDED

SKY BLUE TEE - 5

₹809

VIEW DETAILS



AI RECOMMENDED

SKY BLUE TEE - 6

₹824

VIEW DETAILS



AI RECOMMENDED

SKY BLUE TEE - 7

₹839

VIEW DETAILS



AI RECOMMENDED

SKY BLUE TEE - 8

₹854

VIEW DETAILS

Black T-Shirts Collection



AI RECOMMENDED

PREMIUM BLACK TEE - 1

₹699

VIEW DETAILS



AI RECOMMENDED

PREMIUM BLACK TEE - 2

₹719

VIEW DETAILS



AI RECOMMENDED

PREMIUM BLACK TEE - 3

₹739

VIEW DETAILS



AI RECOMMENDED

PREMIUM BLACK TEE - 4

₹759

VIEW DETAILS

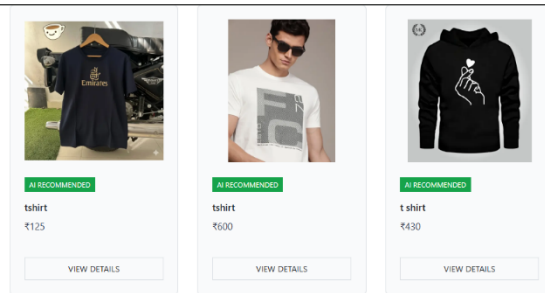


localhost:5173/black









Get connected with us on social networks!



SmartOrder AI

SmartOrder AI is an intelligent e-commerce platform that uses AI-driven decision intelligence to reduce impulse buying and improve customer experience.

Products

Blue Collection
Black Collection
White Collection
Shopping Cart

Useful Links

About Us
Contact Us
Admin Login
Dashboard

Contact

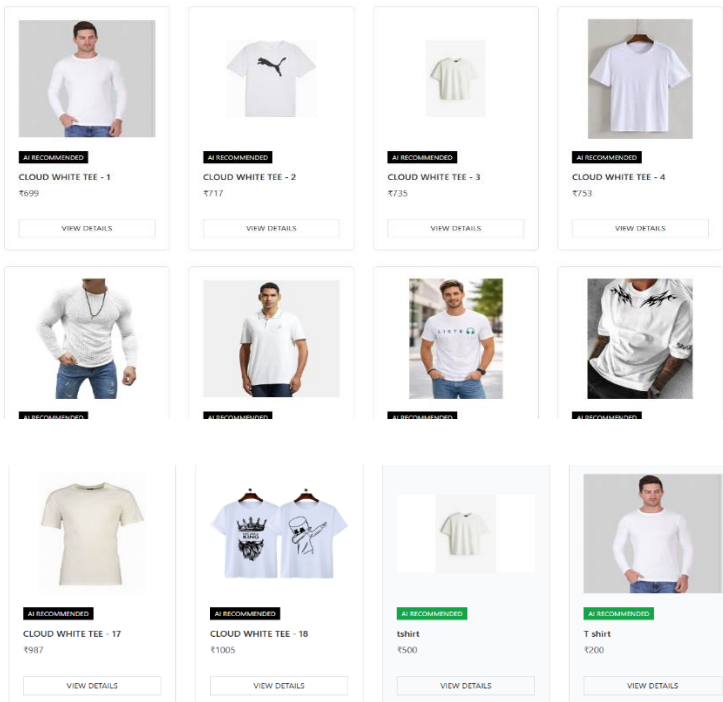
Ahmedabad, Gujarat, India
support@smartorderai.com
+91 9876543210
AI Decision Support System

© 2026 SmartOrder AI | All Rights Reserved

SmartOrder AI
AI-ASSISTED DECISION INTELLIGENCE

SHOP Blue Black White About Contact AI CART Krunal Chauhan CONTROL HUB LOGOUT

White T-Shirts Collection



Get connected with us on social networks!



SmartOrder AI

SmartOrder AI is an intelligent e-commerce platform that uses AI-driven decision intelligence to reduce impulse buying and improve customer experience.

Products

Blue Collection
Black Collection
White Collection
Shopping Cart

Useful Links

About Us
Contact Us
Admin Login
Dashboard

Contact

Ahmedabad, Gujarat, India
support@smartorderai.com
+91 9876543210
AI Decision Support System

© 2026 SmartOrder AI | All Rights Reserved



❖ ABOUT PAGE: -

SmartOrder AI
AI ASSISTED DECISION INTELLIGENCE

SHOPBlueBlackWhiteAboutContactAICARTKrunal ChauhanCONTROL HUBLOGOUT

About SmartOrder AI

SmartOrder AI is an intelligent e-commerce platform that goes beyond traditional online shopping. We analyze user behavior in real-time and provide AI-driven decision insights to reduce impulse purchases and increase smart buying confidence.

98%
Decision Accuracy

50+
AI Behavior Logs

24/7
Real-Time Analysis

100%
Smart Purchase Tracking

How Our AI Works

1 Behavior Tracking
AI tracks time spent, product interaction, color/size selection, and user engagement.

2 Decision Scoring
System calculates Impulse Score, Compatibility Score, and Final Decision Score.

3 Smart Insight
AI suggests whether purchase is Rational or Impulsive before checkout.

Why Choose SmartOrder AI?

- ✓ AI-powered decision intelligence
- ✓ Impulse purchase detection
- ✓ Real-time analytics dashboard
- ✓ Payment & Order tracking system

Our Mission
To transform e-commerce from simple product selling into intelligent buying guidance powered by AI.

Why Choose SmartOrder AI?

- ✓ AI-powered decision intelligence
- ✓ Impulse purchase detection
- ✓ Real-time analytics dashboard
- ✓ Payment & Order tracking system
- ✓ Secure & scalable MERN architecture

Our Mission
To transform e-commerce from simple product selling into intelligent buying guidance powered by AI.

Our Vision
To build the world's first AI-native shopping platform that eliminates buyer regret and maximizes smart decisions.

🚀 Future Roadmap

Upcoming features include AI-based product recommendation, emotion detection, smart budget tracking, advanced data visualization dashboards, and enterprise AI integration.

Get connected with us on social networks

SmartOrder AI
SmartOrder AI is an intelligent e-commerce platform that uses AI-driven decision intelligence to reduce impulse buying and improve customer experience.

Products
Blue Collection
Black Collection
White Collection
Shopping Cart

Useful Links
About Us
Contact Us
Admin Login
Dashboard

Contact
Ahmedabad, Gujarat, India
support@smartorderai.com
+91 9876543210
AI Decision Support System

© 2026 SmartOrder AI | All Rights Reserved

❖ CONTACT PAGE: -

Contact Information

Address
Ahmedabad, Gujarat, India

Phone
+91 9876543210

Email
support@smartorderai.com

Send Us A Message

First Name

Last Name

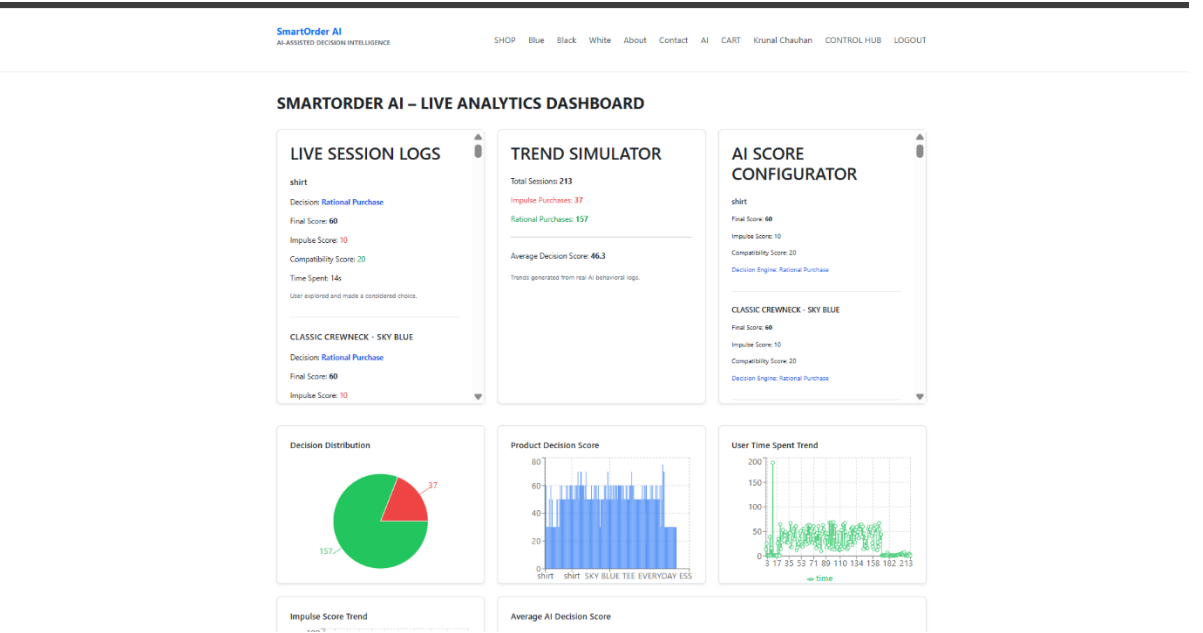
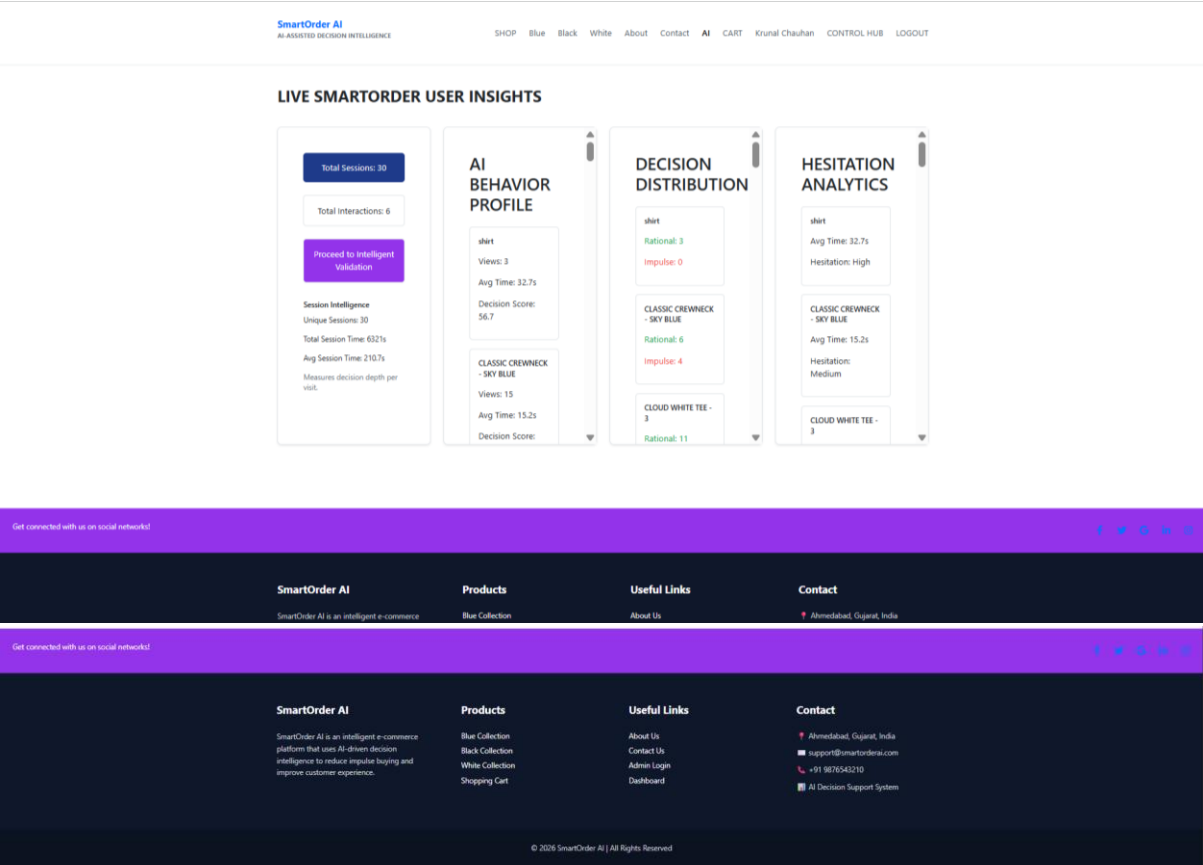
Enter Your Email

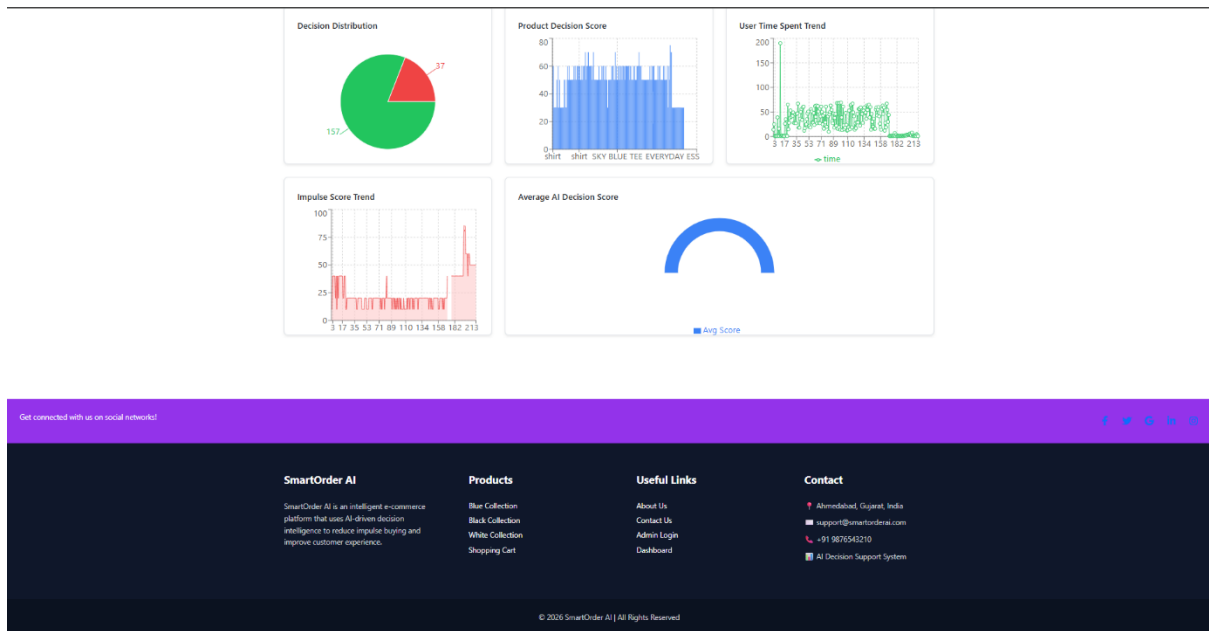
Enter Phone Number

Write Your Message

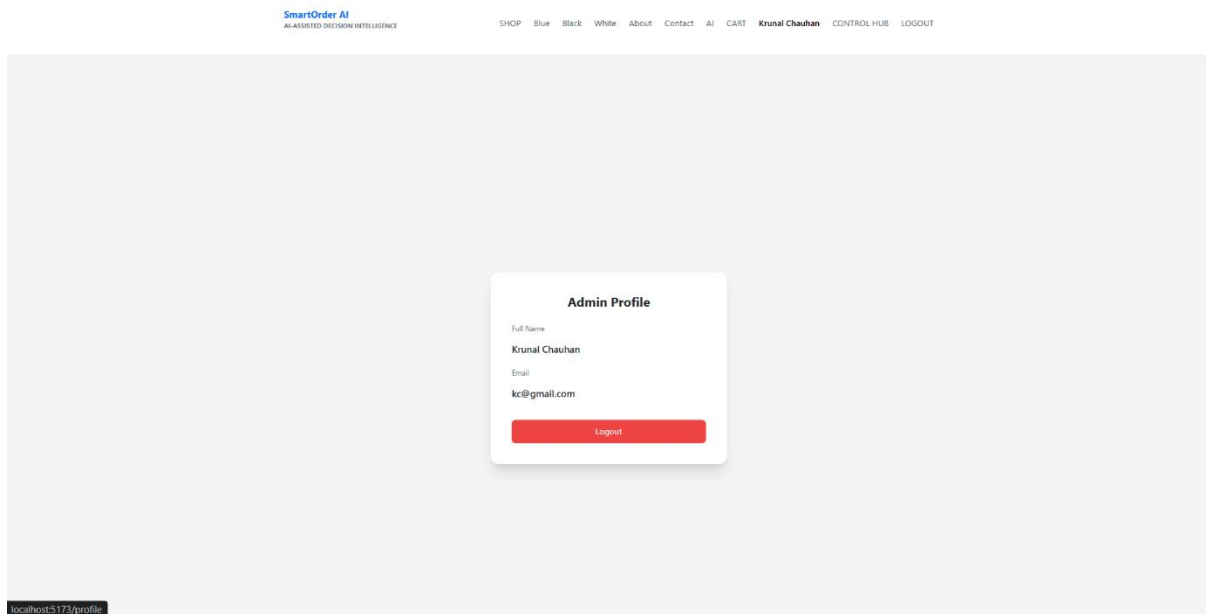
SEND MESSAGE

❖ AI BEHAVIOUR AND TRACKING PAGE: -
This page show change password.





❖ Profile PAGE: -



CHAPTER 10

ADVANTAGES

10.1 Advantages

- The **SmartOrder AI system** provides several advantages compared to traditional e-commerce platforms. The system integrates Artificial Intelligence with an online shopping platform to analyze user behavior and improve the purchasing experience.
- **Improved Decision Making**
 - The system analyzes user interaction data such as time spent on product pages and browsing behavior to determine whether a purchase is rational or impulsive. This helps users make better purchasing decisions.
- **User Behavior Analysis**
 - SmartOrder AI collects behavioral data and analyzes it to generate useful insights about customer decision patterns.
- **Better User Experience**
 - The system provides a modern and responsive interface developed using **React.js and Tailwind CSS**, which improves the user experience.
- **Efficient Product Management**
 - Administrators can easily manage products, orders, payments, and user data through the admin dashboard.
- **Real-Time Analytics**
 - The system provides real-time analytics and AI behavior logs that help administrators understand customer purchasing patterns.
- **Secure Data Management**
 - The system uses **MongoDB database** to securely store user data, product information, orders, and AI logs.
- **Scalable System**
 - The system is developed using **Node.js and React.js**, which allows the application to scale easily when the number of users increases.
- **Modern Technology Stack**
- The project uses modern technologies such as:
 - React.js
 - Tailwind CSS
 - Node.js
 - Express.js
 - MongoDB
- These technologies make the system efficient and easy to maintain.

10.2 Limitations

- Although the SmartOrder AI system provides many advantages, there are some limitations.

- Limited Product Categories
 - Currently, the system focuses only on **T-shirt products**. The system can be expanded in the future to support multiple product categories.
- Basic AI Model
 - The AI behavior analysis in the system is based on simple behavioral metrics. More advanced machine learning models can improve accuracy
- No Advanced Recommendation Engine
 - The system does not currently include a recommendation system based on deep learning or collaborative filtering.
- Dependency on Internet
 - Since the system is web-based, users must have an active internet connection to access the platform.
- Limited Payment Integration
 - The system currently demonstrates order placement but may require integration with secure payment gateways for real-world deployment.

CHAPTER 11

CONCLUSION AND FUTURE ENHANCEMENT

11.1 Conclusion

- The SmartOrder AI project demonstrates how Artificial Intelligence can be integrated into an e-commerce platform to analyze customer behavior and improve decision making.
- The system was developed using modern web technologies including React.js, Node.js, Express.js, MongoDB, HTML, CSS, JavaScript, and Tailwind CSS.
- The platform allows users to browse products, add items to their cart, and place orders through an interactive interface. At the same time, the system collects behavioral data such as time spent on product pages and interaction patterns.
- This data is analyzed using the SmartOrder AI module to classify purchase behavior as Rational Purchase or Impulse Purchase.
- The system also provides an admin dashboard that allows administrators to manage products, monitor orders, track payments, and analyze AI behavior logs.
- The project successfully demonstrates the concept of AI-assisted decision intelligence in e-commerce systems.
- Overall, the SmartOrder AI platform provides a foundation for future intelligent commerce applications and shows how behavioral analytics can improve the online shopping experience.

11.2 Future Enhancement

- Although the SmartOrder AI system works effectively, several improvements can be implemented in the future to enhance the platform.
- Advanced AI Models
 - More advanced machine learning algorithms can be used to improve the accuracy of behavior analysis.
- Product Recommendation System
 - A personalized recommendation system can be added to suggest products based on user preferences and purchase history.
- Multiple Product Categories
 - The platform can be expanded to include additional product categories such as electronics, fashion, and accessories.
- Mobile Application
 - A mobile version of the SmartOrder AI platform can be developed using technologies such as React Native.
- Payment Gateway Integration
 - Integration with secure payment gateways such as Razorpay, Stripe, or PayPal can improve real-world usability.
- Advanced Analytics Dashboard

- More detailed analytics dashboards can be developed for administrators to analyze user behavior and business performance.
- User Profile Personalization
 - The system can provide personalized product suggestions based on user browsing history and preferences.

CHAPTER 12

BIBLIOGRAPHY

12.1 Web Reference

- <http://www.W3schools.com>
- <http://www.brainybeam.com>
- <https://www.slideshare.net/darshan0303/online-final-report>
- <https://www.slideshare.net/piyushkhadse589/online-bus-pass-management-system>

*Thank
you!*